Biologically Informed Corpus Design for Annotation-based Language Models of Prokaryotic Pangenomes

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Abstract

The rapid growth of publicly available prokaryotic genomes is driving a shift toward large-scale comparative pangenomics, while simultaneously creating a need for scalable methods to annotate and exploit these increasingly large collections of proteins. In parallel, recent advances in genomic and protein language models suggest that Large Language Models (LLMs) could provide a powerful framework for learning biological patterns directly from large-scale genomic resources. In this context, the PanGAIMiX project was recently initiated as an ANR-funded initiative encompassing several complementary work packages aimed at leveraging advances in AI for pangenome graphs exploration. As part of this concerted effort, PanGBank, a comprehensive collection of species-level prokaryotic pangenomes, was used as the foundation for constructing and analyzing a large-scale annotation corpus for downstream LLM training. By leveraging the wealth of publicly available annotation resources, this work explored the potential of annotation-level language modeling, where biological annotations act as tokens and enable the learning of higher-order functional and evolutionary relationships.

In this context, large-scale workflows were developed to characterize the representative proteins across PanGBank pangenomes and transform them into an LLM-compatible biological corpus, by integrating: (i) homology exploration through protein clustering, (ii) multi-modal annotation, (iii) ESM protein embeddings computation, and (iv) annotation-level tokenization. Our findings revealed the capacity of DIAMOND DeepClust to cluster the full set of 34.6 million representatives across several stringency levels, reducing it to 4.5 million representatives in the no-threshold partition, with criterion-based, annotation-based, and embedding-based analyses supporting the consistency of the resulting clusters. Subsequently, multi-modal annotation with diverse curated resources showed broad but incomplete coverage and strong complementarity between annotation layers. Moreover, ESM embeddings provided complementary insights into alignment-based homology and annotation structure, including cluster coherence in embedding space and structured transitions between annotation-rich and poorly annotated regions. Lastly, analysis of the constructed annotation vocabularies revealed long-tailed natural language-like frequency distributions, multi-token proteins, and heterogeneous sparsity patterns, motivating a multi-modal tokenization strategy able to preserve complementary functional and structural information.

Keywords: comparative pangenomics, pangenome annotation, protein clustering, protein embeddings, large language models, microbial genomics

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List of Abbreviations

PanGAIMiX Pangenome Graphs for AI-driven Microbial genome eXploration

PPanGGOLiN Partitioned PanGenome Graph Of Linked Neighbors

BMM Bernoulli Mixture Model

NEM Neighboring Expectation-Maximization

HGT Horizontal Gene Transfer

RGP Region of Genomic Plasticity

BGC Biosynthetic Gene Cluster

AMR Anti-Microbial Resistance

HMM Hidden Markov Model

MCL Markov CLustering

MSA Multiple Sequence Alignment

GVC Greedy Vertex Cover

GNN Graph Neural Network

LLM Large Language Model

LLLM Large Language of Life Model

gLM genomic Language Model

pLM protein Language Model

Internship Host

The internship took place at the **LABGeM** (Laboratory of Bioinformatics Analyses for Genomics and Metabolism), hosted within the **UMR 8030 Génomique Métabolique** unit of **Genoscope (CEA)**. Genoscope is known as a prominent national sequencing center in France and was historically part of several major initiatives, including the Human Genome Project. The **UMR 8030** unit is the basic research structure of Genoscope, encompassing 6 specialized laboratories.

Among them, the **LABGeM** specializes in microbial genomics and bioinformatics, with a strong focus on large-scale comparative genomics, pangenomics, and the development of computational methods and resources for microbial genomes annotation and analysis. The laboratory develops and maintains several major bioinformatics tools and databases for microbial genome analysis and pangenome construction, most notably the Microscope platform, the PPanGGOLiN software suite, and the PanGBank database.

Furthermore, the internship is part of the ANR-funded **PanGAIMiX** (Pangenome Graphs for AI-driven Microbial genome eXploration) project, initiated in late 2025 as a collaboration between three laboratories: **LABGeM**, **LaMME** (Laboratoire de Mathématiques et Modélisation d'Évry, Université Évry Paris-Saclay), and **MaIAGE** (Mathématiques et Informatique Appliquées du Génome à l'Environnement, INRAE).

1 Introduction

1.1 General Context

The significant increase in the availability of prokaryotic genomes prompted a parallel evolution in the field of comparative genomics to accommodate for this abundance of data. In this context, pangenomes were proposed as a species-level framework to represent the full genomic diversity across all strains of a species [1]. The pangenome is defined as the complete set of genes across all strains of a species. It is commonly partitioned into a *core* genome comprising the genes shared across all strains and an *accessory* genome of the genes restricted to some strains. Alternatively, a less strict definition divides the pangenome into a *persistent* component, consisting of the gene families present in almost all strains, a *shell* component, consisting of families of intermediate frequencies across strains, and a *cloud* genome of the families present at low frequencies in the species [2].

There exists multiple tools and approaches for pangenomes construction and analysis. Classical approaches rely on predefined thresholds to partition the pangenome into its core and accessory components [3, 4, 5]. These include a sequence identity threshold to group genes into gene families and a presence cut-off specifying the proportion of genomes in which a family has to exist in order to be considered as part of the core genome. Other approaches rely on statistical methods, such as Bayesian models [6] and Binomial mixture models [7], to partition the different components of the pangenome. Furthermore, pangenome graphs were suggested to model a sequence-level consensus representation of many genomes by defining an alternative path for each known nucleotide variation [8, 9]. At LABGeM, PPanGGOLiN (Partitioned PanGenome Graph Of Linked Neighbors) was proposed with an innovative methodology making use of different aspects of previous approaches, by integrating both a statistical Bernoulli Mixture Model (BMM) and a gene-level pangenome graph architecture for pangenome partitioning [10]. PPanGGOLiN uses a graph structure based on genetic contiguity, where nodes represent gene families and edges indicate adjacency of the two connected families on a contig. The edges are weighted by the number of genomes in which the relation of contiguity exists. Unlike sequence-level pangenome graphs, the gene-level graph of PPanGGOLiN provides better biological interpretability and the possibility to exploit genomic contexts, such as identifying conserved operons and understanding functional associations in genomic neighborhoods.

Subsequently, PPanGGOLiN was applied to construct the species-level pangenome graphs of all prokaryotic genomes available in GTDB (The Genome Taxonomy Database), which collects all genomes from GenBank and RefSeq, constructs their phylogenies, and clusters them accordingly. This collection of pangenomes spanning thousands of prokaryotic species was used to create the PanGBank database (Mainguy et al, in preparation; https://pangbank.genoscope.cns.fr), providing an important infrastructure to support moving towards microbial evolution exploration and comparative pangenomics at an unprecedented scale.

Beyond addressing fundamental evolutionary questions, large-scale pangenomic analyses are highly relevant to critical biomedical challenges, metabolic engineering, and synthetic biology applications. For instance, a deep and scalable understanding of prokaryotic pangenomes is essential for studying antimicrobial resistance (AMR), uncovering novel biosynthetic gene clusters (BGCs) for biotechnology, and identifying therapeutic targets against emerging pathogens.

Moreover, the study of prokaryotes has broader implications for global biogeochemical cycles and the human microbiome.

In this context, several comparative pangenomics approaches have been introduced over the years. Most notably, graph-based pangenomics methods have been leveraged to characterize microbial diversity, horizontal gene transfer (HGT), and genomic organization. For instance, panRGP [11] and panModule [12] enabled the exploration of conserved genomic organization and regions of genomic plasticity (RGP) across thousands of genomes. However, notwithstanding their accuracy and usefulness, the computational complexity of such approaches makes them unsuitable for scaling up to hundreds of pangenomes. Moreover, PANORAMA [13] was developed to detect systems and analyze genomic contexts across pangenomes. Yet, its reliance on predefined rule-based models makes it unable of discovering new contextual patterns. Structure-based searches were also proposed as a means to improve sensitivity in identifying remotely related conserved gene clusters, as in the recent Spacedust method [14]. However, requiring all-against-all comparisons significantly limits the scalability and interpretability of such approaches. Overall, previous approaches remain primarily descriptive and rely on explicit statistical or graph-based analyses that do not scale efficiently toward the extraction of higher-order biological semantics from increasingly massive genomic datasets.

In parallel, the recent rapid advances in artificial intelligence approaches offer a promising avenue for overcoming the limitations of traditional methods at the current scale of available prokaryotic genomes. Accordingly, the ANR-funded PanGAIMiX (Pangenome Graphs for AI-driven Microbial genome eXploration) project was recently proposed to introduce several innovative methods that integrate pangenome graph models with state-of-the-art AI approaches. The project is intended to serve as a coordinative framework encompassing four parallel work packages (WPs). Firstly, WP1 aims at upgrading PPanGGOLiN partitioning methodology to allow genus-level pangenomes construction. Secondly, WP2 will rely on graph neural network (GNN) architectures to detect conserved modules across thousands of pangenomes. Thirdly, WP3 aims at training large language model (LLM) transformer architectures on the corpus of functional and structural annotations across all PanGBank pangenomes to enable the prediction of missing annotations and novel biological processes. Fourthly, WP4 intends to leverage PanGBank as a comprehensive open source collection of annotated pangenomes, and conduct two empirical investigations on antibiotic resistance and the discovery of new metabolic pathways. Thus, PanGAIMiX will capitalize on the previously developed PPanGGOLiN and PanGBank resources and combine structure-free and explicit structure-based machine learning approaches to move towards microbial genomes and evolutionary dynamics exploration at an unprecedented scale.

1.2 Motivation & Literature Review

The proposal to extend the recent advances in natural language processing (NLP) and LLMs to microbial genomics studies is rooted in several motivations. First, microbial gene distributions across genomes present compelling natural language-like properties, making them prime candidates for LLM-like modeling. Indeed, there are arguable syntactic and semantic parallels between the evolutionary dynamics of natural languages and microbial gene function and arrangement. Moreover, the vertical and horizontal transfer dynamics that govern the evolution of microbial genomes make them in alignment with the principles of distributional semantics [15] at the basis of LLMs [16]. Second, there is a long-standing tradition of using gene arrangement to help predict gene function and functional associations, commonly referred to as "guilt by association," which presents a direct genomic analogy to word context [17]. Third, prokaryotic genomes

exhibit several properties that facilitate their LLM-like modeling, including dense functional regions, operon-based gene organization, relatively few repetitive elements, and an abundance of available annotations. Fourth, the notable successes of sequence-level genomic and protein LLMs, such as ESM [18, 19] and Evo [20, 21], have demonstrated the utility of applying language models to biology, giving rise to a surge of so-called Large Language of Life Models (LLLMs) [22].

These attempts at leveraging NLP principles for LLLM design have progressively evolved to span multiple levels of consideration. The initial models were genomic local DNA-level (gLM), such as DNABERT and DNABERT2 [23, 24], adapting NLP bidirectional transformer architectures and demonstrating strong performance on sequence-level prediction tasks. Protein-level models (pLM) rapidly followed, most notably ESM [18, 19] and ProtT5 [25]. More recently, genome-level models, such as Bacformer [26], were introduced to account for higher-order genome organization such as operons, gene neighborhoods, and long-range genomic dependencies. Furthermore, Evo adopted an auto-regressive transformer architecture to allow consideration of even larger genomic contexts [20, 21]. Some models are further fine-tuned for specific applications, such as GenSLMs [27] specialized for SARS-CoV-2 genomes. Notably, a few attempts at higher order genomic contexts, such as gene annotations, have been recently explored as well. For instance, Miller et al. modeled gene families as words with contigs as sentences and used the word2vec algorithm [28] to map families into a high-dimensional vector space of contextual embeddings, allowing the inference of gene functions based on their genomic neighborhoods [29]. The approach was further developed into a web tool GeNLP [30]. Another attempt by Hwang et al. stacked a gLM on top of a pre-trained pLM to capture both the intrinsic sequence data and broader genomic context of proteins, demonstrating significant improvement in the prediction of gene functions, metabolic pathways, and operons [31].

Notwithstanding their successes, sequence-level gLMs and pLMs come with tremendous computational and financial costs, as the attention of transformer-based LLM architectures scales quadratically with sequence length [32]. Moreover, these purely alignment-based models neglect the valuable substantial amount of curated biological knowledge already available in databases. In addition, working with raw DNA requires handling redundant and weakly informative regions. On the other hand, the previous attempts to work with higher order genomic patterns tended to rely on strict pre-defined gene families and use simple architectures with limited capacity to capture long-range contextual dependencies. To overcome all these limitations, we aim to move towards an annotation-level LLLM with an encoder-decoder BART/T5 architecture. This approach secures more than a thousand-fold reduction in computational costs by abstracting genomes into functional units, while unlocking several underexplored advantages through the auto-regressive decoder phase, such as generative functional assignment and the synthetic design of co-regulated genomic architectures like operons. Moreover, WP3 proposal aims to work with function-level tokens that would integrate heterogeneous biological annotations, spanning Pfam domains, orthologous groups, metabolic pathways, and biological processes. Thus, another added value of this approach would be its capacity for multi-modal translation based on cross-attention training mechanisms, as well as multi-level biological reasoning from domain to pathways and biological processes.

In this context, the proposal to train an LLM on PanGBank pangenomes first requires a comprehensive characterization and preprocessing of this underlying dataset, which will serve as the model's training corpus. This primarily involves assessing homology across all gene families to define the basic functional units across all pangenomes, as well as performing large-scale

multi-modal annotation to integrate complementary information on their functional roles, protein domains, orthologous group assignments, and structural superfamily classifications. Several large-scale resources have been developed for related purposes, demonstrating the value of systematic integrative annotation for transforming heterogeneous microbial genomic data into structured reusable knowledge bases. InterProScan6 provides a modern framework for large-scale protein function annotation, integrating InterPro member-database signatures and associated prediction tools to support broad functional characterization [33]. MGnify provides a well-established resource for microbiome sequence analysis, in which public metagenomic and metatranscriptomic datasets are processed into taxonomic and functional annotation layers [34]. More recently, mettannotator has been introduced as a scalable Nextflow pipeline for exhaustive prokaryotic genome annotation and has been among the primary annotation workflows supporting MGnify [35]. In the prokaryotic genome space, proGenomes4 provides nearly two million consistently annotated bacterial and archaeal genomes with standardized taxonomic, ecological, and functional metadata [36]. Together, these efforts motivate a comparable preprocessing strategy for PanGBank, in which pangenome-derived gene families are converted into a comprehensive annotation-rich corpus suitable for downstream LLM training.

In parallel, several frameworks for large-scale clustering of microbial coding sequences have been introduced over the years. AGNOSTOS demonstrated how known and unknown microbial coding sequence space can be unified through systematic gene clustering, enabling the characterization of both annotated and poorly characterized gene families across genomes and metagenomes [37]. More recently, DIAMOND DeepClust extended this problem to the scale of the global protein universe, providing an ultra-fast strategy for clustering billions of protein sequences and reducing the dimensionality of large protein collections [38]. Complementarily, the nf-core/proteinfamilies workflow, adopted by MGnify, allows generating or updating protein families from a set of sequences, producing clustered protein families together with multiple sequence alignments and profile HMMs [39]. Accordingly, a similar scalable clustering step is necessary for PanGBank preprocessing, whereby homologous genes are grouped into coherent family-level units prior to downstream analyses and training.

Beyond alignment-based clustering and curated annotations, pLM embeddings provide an additional representation layer for characterizing large protein collections. Indeed, foundational work on ESM showed that unsupervised language models trained on large protein sequence corpora learn representations that encode structural and functional properties of proteins [18]. Several recent studies have further shown that ESM-derived embeddings can support alignment-free or weakly supervised clustering of protein families and evolutionary groups. For instance, Yeung et al. demonstrated that tree visualizations of protein embedding space improve functional clustering of diverse protein superfamilies [40], while Jiao et al. explored graph-enhanced sequence modeling and efficient clustering based on ESM2 representations [41]. In addition to clustering, protein embeddings can support the transfer of annotations to poorly characterized sequences by capturing functional and structural similarities that are undetectable by traditional HMM- and alignment-based annotation approaches [42]. PLMSearch illustrates this potential for remote homology detection, successfully identifying remote homology pairs with dissimilar sequences but similar structures [43]. Moreover, recent applications to viral protein annotation and GO-based function prediction show that protein language model representations can help assign candidate functions to weakly annotated proteins [44, 45]. Overall, these recent findings emphasize the importance of computing ESM embeddings for PanGBank representative proteins to provide a useful complement to conventional homology- and domain-based annotations, enabling the recovery of remote functional relationships.

1.3 Internship Objectives

This internship primarily aims to prepare PanGBank pangenomes for downstream LLM training through the following tasks:

(1) Clustering representative proteins across all PanGBank pangenomes

(2) Designing and executing a large-scale multi-modal annotation workflow

(3) Computing protein embeddings for the compressed set of representative sequences

(4) Developing a genomic tokenizer to map the multi-layered annotations into a corpus of LLM-compatible tokens

The clustering step is aimed at exploring alignment-based homology at multiple granularity levels across the gene families of all PanGBank pangenomes. It identifies the basic representative groups that capture the diversity of the dataset at multiple levels of sequence similarity. Accordingly, it enables the reduction of the dataset scale by only considering the final cluster representatives for downstream computations. In parallel, the computation of protein embeddings provides a complementary view of homology that overcomes the intrinsic alignment-based and threshold-dependent limitations of traditional clustering approaches. Indeed, embeddings capture structural and evolutionary patterns that transcend direct sequence similarity. This makes them particularly valuable for remote homology detection, especially within the twilight zone of sequence similarity, motivating their use for cluster refinement, the construction of higher-order superclusters from alignment-based clusters, and annotation transfer to reduce missing annotations—an important challenge for downstream LLM training. Moreover, protein embeddings could be directly incorporated into LLM architectures, for instance by serving as a biologically-informed initialization of the model's embedding layer instead of the typical random initialization.

Together, alignment-based clusters, multi-modal annotations, and protein embeddings provide three valuable complementary sources of insights that collectively enable a comprehensive characterization of the PanGBank dataset. Accordingly, the overlap and divergence between these approaches will be evaluated, and their respective strengths will be leveraged to construct a high-quality, biologically informed corpus for the downstream LLM training.

Finally, besides their direct relevance for PanGAIMiX WP3, the internship objectives overlap with the three other work packages. The clustering step is directly relevant to WP1, which aims to construct genus-level pangenomes, thus requiring moving from the current species-level gene families in PanGBank towards exploring higher-level families with less stringent homology. In parallel, the computation of protein embeddings was done in direct collaboration with WP2, which relies on embedding-based similarity to detect homology and link protein nodes across pangenome graphs prior to GNN training. Lastly, the developed clustering and annotation workflows, as well as all obtained clusters, annotations, and protein embeddings will be integrated into PanGBank as part of WP4, to leverage it as a centralized infrastructure facilitating collaboration across work packages and providing access to the broader community.

2 Materials & Methods

2.1 PanGBank

All annotation, clustering, and embedding computations were developed and evaluated on PanGBank pangenomes. PanGBank (https://pangbank.genoscope.cns.fr/; Mainguy et al, in preparation) is a database of species-level pangenomes precomputed with PPanGGOLiN for all publicly available prokaryotic species represented in the GTDB (Genome Taxonomy Database), a standardized genome-based taxonomy for Bacteria and Archaea [46]. It includes two collections: one comprising all GTDB genomes sourced from NCBI GenBank (Genetic Sequence Database) and RefSeq (Reference Sequence Database), and another restricted to RefSeq genomes. This separation is motivated by the better annotation quality of RefSeq, whereas including all genomes enables the construction of a larger number of pangenomes, at the cost of re-annotating all genomes to avoid annotation heterogeneity. Moreover, owing to the minimum 15-genome requirement for PPanGGOLiN pangenome construction, only 4351 of the total 143,614 GTDB species are retained in PanGBank. The RefSeq-only collection is further reduced to 1770 pangenomes. Moreover, to keep pangenome sizes computationally manageable, PanGBank limits each species pangenome to a maximum of 2000 genomes. All pangenomes and associated metadata from both PanGBank collections are accessible through an API and a web interface. The scale of both collections is summarized in Table 1. Clustering, annotation, and embeddings were computed at the scale of PanGBank representative proteins, which for each pangenome, correspond to its set of protein families inferred as part of PPanGGOLiN pangenome construction through an MMseqs clustering of the repertoire of all species genes at 80% sequence identity.

Collection	Genera	Pangenomes	Genomes	Representative Sequences	Total Genes
GTDB-all	1,422	4,351	334,252	34,622,399	1,032,248,203
GTDB-RefSeq	443	1,770	183,331	15,740,412	469,186,290

Table 1: Scale of the PanGBank collections (v1.0.0) used for clustering, annotation, and embedding analyses. The representatives count highlights the total number of families obtained as part of PPanGGOLiN pangenome construction workflow, through an MMseqs clustering at 80% identity applied to the repertoire of all genes per pangenome.

2.2 Clustering

2.2.1 Clustering approaches

MMseqs MMseqs2 (Many-against-Many sequence searching) is a bioinformatic software suite for sequence similarity searches and homology-based clustering [47]. It bypasses the quadratic time complexity $O(N^2)$ of traditional all-vs-all alignments by utilizing a highly optimized, k-mer-based pre-filtering index that scales linearly with database size. Its clustering workflow conveniently combines the pre-filtering, alignment and clustering modules into either a simple or cascaded clustering of a sequence database.

nf-core/proteinfamilies The *nf-core/proteinfamilies* pipeline is a recent clustering workflow [39], developed under the nf-core framework, a prominent initiative hosting a collection of standardized peer-reviewed bioinformatics pipelines built with Nextflow [48]. The workflow starts by computing MMseqs clusters, before applying a series of HMM-based refinement steps (Supplementary Fig. S1). Its core idea is to compute multiple sequence alignments (MSA) for each cluster exceeding a user-defined size threshold to proceed with building cluster-specific HMM profiles, which are then deployed via *hmmsearch* to iteratively recruit distant homologous sequences that pass significant statistical thresholds. The primary motivation is that the position-specific scoring matrices inherent to HMM profiles offer substantially higher sensitivity than traditional alignment scores, allowing the pipeline to better resolve remote homologies in the evolutionary twilight zone. Finally, the workflow offers optional downstream steps to remove intra-cluster redundancies through an additional round of MMseqs in *linclust* mode.

DeepClust DIAMOND is a sequence alignment tool designed to query sequences against massive databases thousands of times faster than traditional tools like BLASTp, while maintaining a comparable level of sensitivity [49]. Building upon this efficient alignment approach, DIAMOND clustering starts with an all-vs-all DIAMOND BLASTp search, filters out weak alignments based on the alignment scores, sequence identity percentages, and E-values of matching pairs, constructs a massive sequence-similarity network from the surviving alignments, and applies the greedy vertex cover algorithm to partition the network, assigning the centroid of each cluster as its representative sequence. More recently, DIAMOND DeepClust was developed to allow efficient clustering at the increasingly immense scale of available sequences [50, 38], with a proof-of-concept application to an unprecedented set of more than 19 billion sequences from across the tree of life. DeepClust's main innovation is its ultra-fast empirically-tuned cascaded clustering approach with multiple clustering rounds of increasing sensitivity. At the end of each round, only the representatives of the obtained clusters are retained for the next stage. This ensures that the more computationally expensive sensitive steps are applied to progressively smaller sets of proteins.

Clustering Parameters The clustering criteria, used for the similarity network construction and partitioning, are defined as:

(1) BLASTp or MMseqs alignment E-value
(2) unidirectional member-to-representative coverage percentage
(3) sequence identity percentage

To capture homology at multiple levels, clustering was performed at 30, 50, and 80% identity thresholds on the full set of representative proteins aggregated across the pangenomes of each PanGBank collection, separately. DeepClust was also executed with its default 0% identity threshold, capturing homology beyond the twilight zone. Coverage was set to an 80% cut-off, relative to the longer sequence, in order to prevent multi-domain proteins from falsely grouping together based on isolated local motifs. The E-value threshold was set to 10^{-5}. For DeepClust cascaded clustering, five consecutive rounds were executed with increasingly more sensitive all-vs-all BLASTp runs, starting from a fast linear mode and ending with the very-sensitive BLASTp mode. All other parameters were left to their defaults for each tool.

Partition Comparison Metrics Agreement between the partitions obtained by different approaches was assessed using:

(1) **Best-match Jaccard index**: it measures how well each cluster from one partition is recovered by its closest counterpart in the other partition. For a cluster C_i from one partition and a second partition V; it is defined as:

$$J_{\text{best}}(C_i, V) = \max_{D_j \in V} \frac{|C_i \cap D_j|}{|C_i \cup D_j|},$$

where D_j ranges over all clusters in V.

(2) **Adjusted Rand index (ARI)**: for two partitions U and V, with $n_{ij} = |U_i \cap V_j|$, row sums a_i, column sums b_j, and total number of proteins n, ARI was computed as:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}.$$

ARI values close to 1 indicate strong agreement, while values close to 0 indicate agreement close to random expectation.

(3) **Normalized mutual information (NMI)**: it measures the amount of information shared between two partitions:

$$\text{NMI}(U, V) = \frac{2I(U;V)}{H(U) + H(V)},$$

where $I(U;V)$ is the mutual information between partitions and $H(U)$, $H(V)$ are their entropies. Values close to 1 indicate strong global correspondence.

2.2.2 Cluster Validation

Three distinct validation approaches were explored to assess cluster quality: (1) criterion-based validation, assessing the extent to which the clustering criterion holds within but not between clusters; (2) annotation-based validation, examining the consistency of functional annotations from different tools among cluster members and their corresponding representative; and (3) embedding-based validation, measuring the member-to-representative embedding similarity within each cluster.

Criterion-based Criterion-based validation assessed the extent to which cluster assignments satisfied the same criteria used during clustering. The following metrics were computed:

1. **Inter-cluster matches**: fraction of representatives that satisfy the clustering criterion with at least one other representative.
2. **Coverage**: fraction of evaluated cluster members with at least one BLASTp hit to any

representative satisfying the clustering criterion:

$$\text{coverage} = \frac{N_{\text{members with hits}}}{N_{\text{evaluated members}}}.$$

3. **Assignment accuracy**: fraction of covered members assigned to their best-supported representative:

$$\text{accuracy}_{\text{hits}} = \frac{N_{\text{correct assignments}}}{N_{\text{members with hits}}}.$$

4. **Overall assignment accuracy**: fraction of all evaluated members assigned to their best-supported representative:

$$\text{accuracy}_{\text{overall}} = \frac{N_{\text{correct assignments}}}{N_{\text{evaluated members}}}.$$

5. **Mean cluster purity**: average assignment accuracy across clusters:

$$\text{Mean cluster purity} = \frac{1}{N}\sum_{c=1}^{N} \text{accuracy}_{\text{hits},c},$$

where N is the number of evaluated clusters.

To compute the described metrics, we define the ground truth using BLASTp in ultra-sensitive mode, between all cluster members and the set of all representatives, as well as among representatives themselves, as highlighted in the clustering pipeline diagram in Fig. 1. The coverage cut-off was set unidirectionally from member to representative, and bidirectionally for representative-vs-representative alignments, with a threshold of 80%. The E-value threshold was set to 10^{-5}, and identity cut-offs were chosen to match the threshold used for each corresponding clustering set.

Functional Consistency Annotation-based validation was assessed by computing the proportion of cluster members whose annotations were consistent with those of their corresponding representative for each annotation tool. For proteins associated with multiple annotations from a given tool, both exact-match and Jaccard similarity metrics were computed to quantify annotation agreement:

$$s_i^{\text{exact}} = \mathbb{I}\left(A_{m_i} = A_{r_i}\right)$$

$$s_i^{\text{Jaccard}} = \frac{\left|A_{m_i} \cap A_{r_i}\right|}{\left|A_{m_i} \cup A_{r_i}\right|}$$

where A_{m_i} and A_{r_i} denote the annotation sets of member i and its representative, respectively.

Missing annotations were handled with different strategies: either considering two empty annotation sets as equivalent or excluding such cases from the analysis. The most conservative estimate was obtained by restricting the computation to per-cluster pairs for which annotations

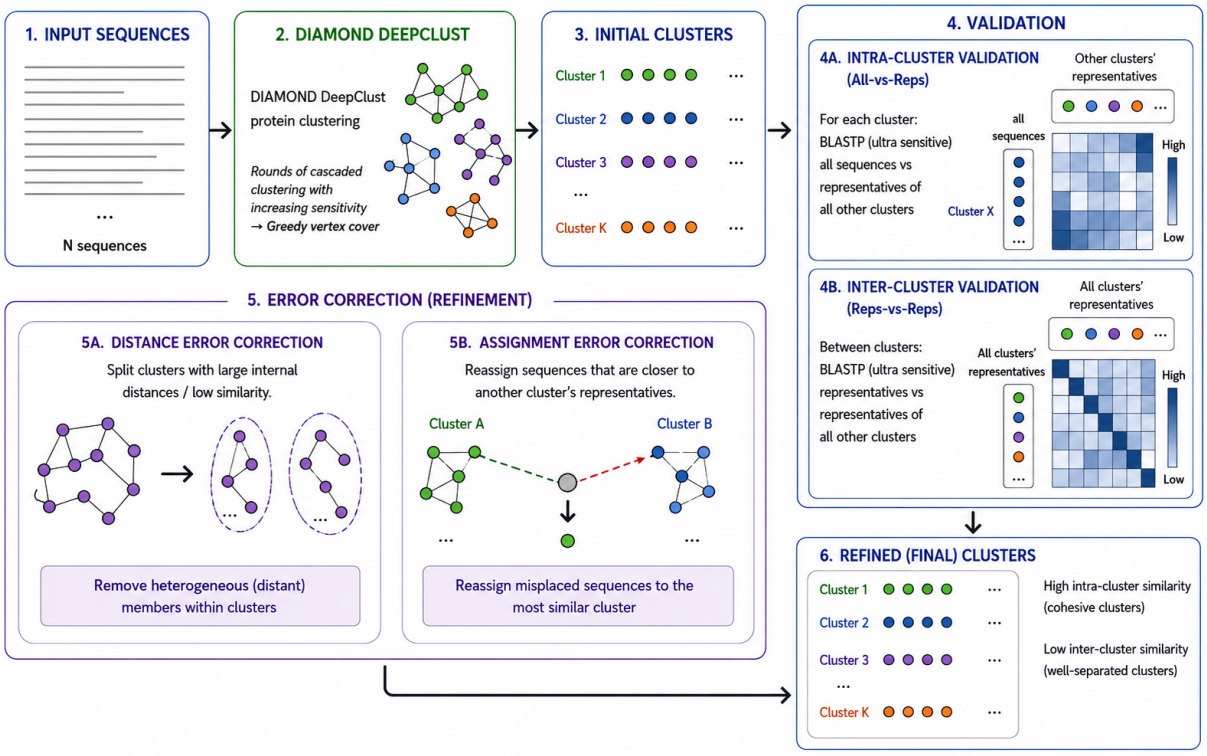


Figure 1: Overview of the clustering, validation, and refinement workflow applied to the aggregated PanGBank representative protein sequences. Initial sequence clusters are generated with DIAMOND DeepClust or MMseqs2, evaluated against alignment-, annotation-, and embedding-based consistency criteria, and optionally corrected through representative reassignment or cluster splitting.

were available for both the member and its representative.

$$\text{Agreement} = \frac{1}{|S|} \sum_{i \in S} s_i,$$

where s_i is either the exact-match or Jaccard score and S is either:

$$S_{\text{all}} = \{\text{all evaluated member–representative pairs}\},$$

Or

$$S_{\text{both}} = \left\{ i : A_{m_i} \neq \varnothing \text{ and } A_{r_i} \neq \varnothing \right\}$$

Embedding-based To assess the cohesion of DIAMOND DeepClust clusters in the protein embedding space, 20,000 clusters containing at least 15 members were selected, and up to 100 proteins were sampled from each of them. ESM embeddings were then computed for the sampled members, and the cosine similarity between each member and its corresponding representative was measured, after applying centering transformation, L2 normalization, and mean pooling (see Section 2.4). In parallel, pairwise cosine similarities were computed between all cluster representatives and used as a null distribution. The significance of member-to-representative similarity in the embedding space was subsequently assessed relative to this null distribution.

2.2.3 Cluster Refinement

Two complementary correction strategies were used to reduce the error metrics defined above, as illustrated in the clustering pipeline diagram (Fig. 1). This was achieved through the *reassign* and *recluster* workflows implemented in DIAMOND. For a given clustering, *reassign* attempts to reassign each non-representative sequence to the closest representative satisfying the clustering criterion, using the e-value of the local alignment as the primary similarity measure. In contrast, *recluster* recomputes a clustering from an existing one while enforcing that all cluster members satisfy the clustering criterion with their representative. As a result, heterogeneous clusters may be split into multiple smaller clusters, increasing the total number of clusters while improving their internal consistency. Both workflows rely on multiple rounds of BLASTp searches performed at different sensitivity levels, allowing problematic assignments in the original clustering to be identified and corrected efficiently.

Alternatively, the potential of Markov CLustering (MCL) to improve the criterion-based quality of the clusters was also investigated. MCL is a graph-based clustering algorithm that simulates random walks on a similarity graph and identifies densely connected regions as clusters [51]. Notably, it enables control over the granularity of the resulting clusters through its inflation parameter (-I), with higher values generally producing finer clusters.

2.3 Annotation

A comprehensive multi-modal annotation pipeline was implemented in Nextflow [52] following the nf-core [48] template and guidelines. It integrates the following annotation resources: Pfam [53], NCBIfam [54], CATH [55], SUPERFAMILY [56], EggNOG [57], KOfam [58], and AMRFinderPlus [59]. A representative diagram of the workflow is shown in Fig. 2. In addition, the clustering pipeline was added as an optional pre-processing step at the start of the workflow. The workflow starts by retrieving all pangenome inputs through PanGBank API and integrates its resulting clusters and annotations within the database at the end.

Conserved domains Pfam provides curated protein domain families based on HMM profiles, enabling sensitive detection of conserved functional and structural units. NCBIfam extends this concept by offering a large collection of protein families derived from RefSeq, supporting broad-scale annotation consistency across bacterial and archaeal genomes.

Structural superfamilies CATH classifies protein domains based on structural and evolutionary relationships, providing a structure-informed view of protein function. Similarly, SUPERFAMILY assigns proteins to structural superfamilies using HMMs derived from known protein structures, enabling structural inference even in the absence of close sequence homologs.

Functional classes In parallel, KOfam provides KEGG Orthology assignments using HMM profiles, linking proteins to metabolic pathways and higher-level functional modules. AMRFinderPlus specializes in the detection of antimicrobial resistance genes and related stress-response determinants, providing clinically relevant functional annotation for pathogen surveillance and resistance tracking with relevance to downstream study cases.

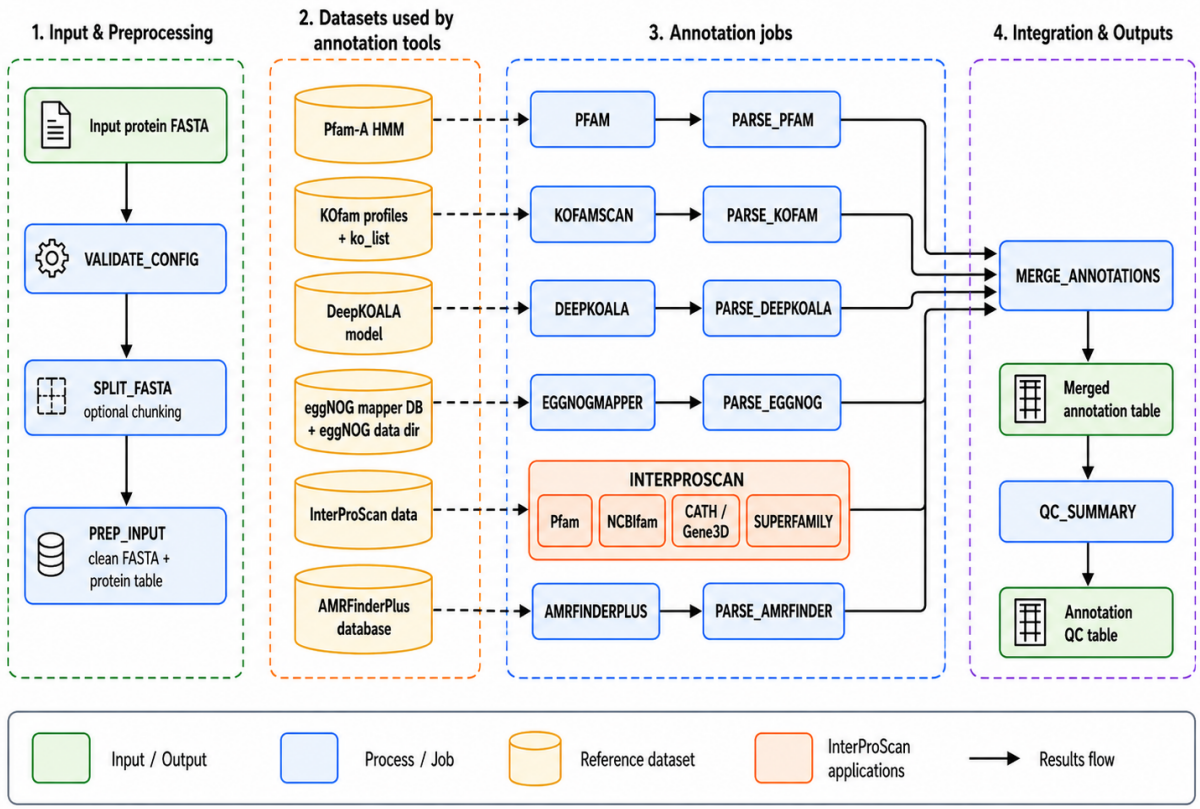


Figure 2: Architecture of the protein annotation workflow implemented in Nextflow. The pipeline processes PanGBank representative protein sequences, optionally after clustering, and integrates complementary functional and structural annotations from InterProScan, EggNOG-mapper, KOfamScan, DeepKOALA, and AMRFinderPlus.

Orthologous groups EggNOG groups proteins into orthologous groups across a wide range of taxa, facilitating functional transfer and evolutionary context inference. While EggNOG integrates multiple annotation types, including KO terms and Pfam domains, the root orthologous group term was mainly retained for it.

Annotation parameters InterProScan6 framework was used to obtain Pfam, NCBIfam, CATH, and SUPERFAMILY annotations with default parameters [60]. To optimize computation, the InterPro precalculated match lookup service was enabled, allowing annotations to be retrieved from precomputed results and avoiding de novo computations where possible. EggNOG-mapper (v2.1.13) was used to obtain EggNOG annotations based on the eggNOG database v5.0.2 [61]. For KO terms, both KOfamScan and the more recent deep learning–enhanced DeepKOALA tool [62] were used. The filtering criteria applied to each annotation resource are summarized in Table 2.

2.4 Protein Embeddings

Protein embeddings were computed with ESM2, a large transformer-based pLM trained on hundreds of millions of protein sequences to generate high-dimensional numerical representations that capture sequence, structural, and functional properties [18, 19]. ProtT5 was also tested to a lesser extent for comparison [25].

Table 2: Filtering criteria used for each annotation resource.

Tool	Filtering criteria
Pfam	Curated gathering cutoff (`-cut_ga`)
NCBIfam	Curated trusted cutoff (`-cut_tc`)
CATH-Gene3D	E-value $\leq 10^{-3}$, coverage $\geq 80\%$, and bitscore ≥ 25
SUPERFAMILY	E-value $\leq 10^{-4}$
EggNOG-mapper	DIAMOND sensitive mode with E-value $\leq 10^{-3}$
KOfamScan	KO-specific cutoffs and hit E-value $< 10^{-5}$
DeepKOALA	Top-hit KO probability $> \sim 0.5$

Both *ESM2_t30_150M_UR50D* and *ESM2_t33_650M* models were evaluated, using their pre-trained models available on the Hugging Face platform (https://huggingface.co/), a widely used distribution hub for AI models and datasets. Notably, ESM2 models scale up to 48 transformer layers and 15 billion parameters, yet we prioritized the 150M- and 650M-parameter variants as they provide a balanced trade-off between representation quality and computational costs.

As raw ESM embeddings are generated at the amino-acid level, they result in a variable-length matrix representation for each protein. A subsequent pooling step is, thus, required to obtain a consistent fixed-size protein-level embedding suitable for downstream analyses. Accordingly, several pooling strategies were explored, including max pooling, mean pooling, CLS-token pooling, and Pool-PaRTI [63]. While max pooling retains the maximum value observed in each embedding dimension, CLS pooling uses the special classification token prepended to each sequence by the transformer architecture, and Pool-PaRTI [63] uses PageRank, a graph-based centrality algorithm, to infers residue importance from attention-derived interaction graphs. Mean pooling was primarily adopted in out computations as it proved to produce final embeddings very similar to those of Pool-PaRTI with a much simpler formula:

$$\mathbf{e} = \frac{1}{L} \sum_{i=1}^{L} \mathbf{h}_i$$

where $\mathbf{h}_1, \ldots, \mathbf{h}_L$ denote the residue-level embeddings of a protein of length L.

Moreover, the centering transformation was applied to handle the anisotropy inherent to ESM embedding space, and consistent with several previous reports [64, 65]. Anisotropy refers to the tendency of ESM space to be dominated by a few directions rather than being uniformly distributed across all dimensions, leading to inflated similarity values between unrelated sequences and other undesirable artifacts. Pooled embeddings were centered using the mean of all embeddings $\boldsymbol{\mu}$:

$$\mathbf{e}_c = \mathbf{e} - \boldsymbol{\mu}; \qquad \boldsymbol{\mu} = \frac{1}{N} \sum_{j=1}^{N} \mathbf{e}_j$$

Finally, L2 normalization was applied as: $\hat{\mathbf{e}} = \mathbf{e}_c / \|\mathbf{e}_c\|_2$.

Annotations in embedding space Annotations were examined in the embedding space by projecting the embeddings of all representative sequences onto a 2D landscape with Incremental PCA. Alternatively, UMAP and t-SNE were also explored. Subsequently, the spatial organization of the different annotation categories and the extent to which they form coherent regions in the embedding space were explored. Inspired by Prabakaran & Bromberg [64], shuffled protein sequences derived from the ASTRAL40 database [66] were included as negative controls. By preserving amino acid composition while disrupting biological sequence organization, these controls are meant to assess the extent of overlap between random sequences and unannotated regions of the embedding space.

ESM-based clustering In order to leverage ESM embeddings for alignment- and thresholds-free clustering, L2-normalized centered embeddings were used to construct an approximate nearest-neighbor (kNN) graph based on cosine similarity (Supplementary Fig. S3). Each node was linked to its 30 closest neighbors (k=30) and weak edges (cosine sim < 0.3) were filtered out. The filtered graph was subsequently clustered with the Leiden community detection algorithm [67].

For structural validation, a panel of 3000 representatives was sampled from the 60 non-singleton ESM-derived superclusters, with 50 representatives retained per supercluster. Structures were predicted from amino-acid sequences using ESMFold [19], with each predicted structure assigned a median pLDDT score, where pLDDT is the model's per-residue confidence estimate on a 0–100 scale. Pairwise structural similarity was computed with Foldseek [68] using the conservative score $\min(qTM, tTM)$, where qTM and tTM are measures of fold similarity normalized by the query and target structure lengths, respectively. Taking the minimum of the two reduces inflation from partial or one-sided alignments. Same-supercluster and between-supercluster pairs were compared globally and after filtering by prediction confidence using several median pLDDT thresholds (50, 60, 70, and 80), to test whether structural support strengthened when lower-confidence predictions were excluded. Enrichment of same-supercluster membership among top-ranked structural matches, same-cluster AUC, and nearest-neighbour recovery were used to assess to what extent ESM superclusters captured detectable structural coherence.

2.5 Tokenization

Annotations were transferred from the representatives of PanGBank pangenomes to all constituent genomes, as highlighted in Supplementary Fig. S2. To this end, the PPanGGOLiN *write_genomes* functionality was used to extract the annotated genome tables of each pangenome, resulting in a collection of 334,252 total genomes with more than 1 billion genes from across the taxonomic groups of GTDB (Table 1). As these tables include the family assignment information of each gene, based on the MMseqs clustering performed during PPanGGOLiN pangenome construction, the annotations assigned to representative protein families can be propagated to all genes belonging to the corresponding families. When the annotation workflow is preceded by a global DeepClust clustering step across all PanGBank representative proteins, an equivalent second propagation stage is performed, where annotations are first transferred from DeepClust clusters to PanGBank representatives and subsequently propagated to all genes through the PPanGGOLiN MMseqs-based family assignments.

Subsequently, the obtained genome-level sets of ordered annotations were tokenized for each tool, as shown in Fig. 3. A shadow token encoding taxonomic identity was appended at the start

of each genome to provide explicit taxonomic context and allow for lineage-aware representations during model training.

Figure 3: Overview of the annotation-based tokenization strategy used to transform protein-level annotations into token sequences for downstream language-model training.

Vocabulary Analysis Metrics The Gini coefficient was used to quantify inequality in annotation-token frequencies. A value of 0 indicates a perfectly uniform distribution, whereas a value close to 1 indicates that a small number of tokens account for most occurrences:

$$G = \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2n \sum_{i=1}^{n} x_i}$$

where x_i is the frequency of token i, and n is the number of tokens.

Zipf's law was used to assess whether annotation-token frequencies follow a long-tailed rank-frequency structure similar to natural language, where a small number of frequent tokens coexist with many rare tokens;

$$f(r) \propto \frac{1}{r^{\alpha}}$$

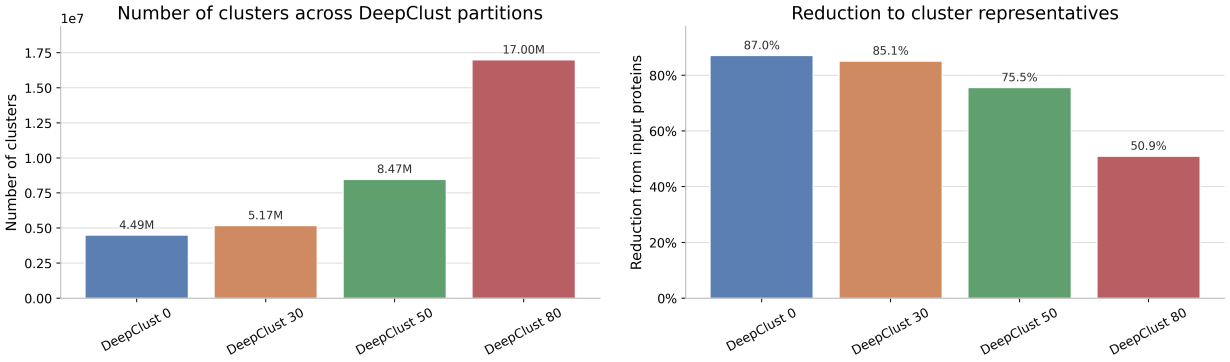
where $f(r)$ is the frequency of the annotation term with rank r, and α is the Zipf exponent. Under the classical Zipf law, $\alpha \approx 1$.

3 Results

3.1 Clustering representative proteins across PanGBank pangenomes

Clustering was applied to the aggregated set of representative proteins across all PanGBank pangenomes to provide a global view of homology within this comprehensive repertoire of proteins spanning distant taxonomic groups. Several clustering approaches were evaluated prior to the full-scale execution, namely MMseqs2, the nf-core/proteinfamilies workflow, and DIAMOND DeepClust (see Section 2.2.1). To determine which approach to adopt, a comparative assessment was performed on a random subset of PanGBank representative proteins. In agreement with the improved scalability reported for DeepClust relative to MMseqs2 [50], DeepClust provided the best runtime in our tests and proved to be the most suitable approach for downstream analyses. In addition to its computational efficiency, it produces a simple and directly exploitable clustering output, with each protein assigned to a single representative cluster. By contrast, the nf-core/proteinfamilies workflow was less adapted to our objectives as its HMM-based refinement and redundancy-removal steps can assign proteins to multiple clusters or remove proteins from the final cluster set. These properties are useful for protein family discovery, but less compatible with our need for a complete, non-overlapping partition of representative proteins for annotation transfer, validation, and corpus construction. The detailed results of this benchmark are provided in Appendix B.

Starting from the approximately 34.6 million representative proteins of the GTDB-all PanGBank collection (Table 1), DeepClust achieved an 87% reduction of the protein space with its least stringent clustering run (0% identity threshold), resulting in 4,492,841 clusters (Fig. 4). Notably, even at 80% identity threshold, DeepClust still achieved a substantial 50% reduction, highlighting the considerable redundancy within the representative proteins repertoire.



(a) Number of DeepClust clusters across identity thresholds.

(b) Reduction of the representative protein set after clustering.

Figure 4: Effect of DeepClust sequence identity stringency on final cluster counts and representative-set reduction across PanGBank proteins.

Furthermore, the proportion of singleton proteins remained stable at around 63% across thresholds, increasing to 73% only for the most stringent clustering at 80% identity (Fig. 5). However, this high percentage of singleton clusters corresponds to a much smaller fraction of the total protein space, as low as 8% (2.8M out of the initial 34.6M) for DeepClust with no identity cut-off. On the other hand, more stringent clustering left increasingly larger fractions of proteins isolated, reaching up to 35% of the total protein repertoire at 80% identity. This high rate of

singletons is expected for large prokaryotic protein collections, where conserved housekeeping families coexist with numerous rare, lineage-specific, or poorly conserved proteins.

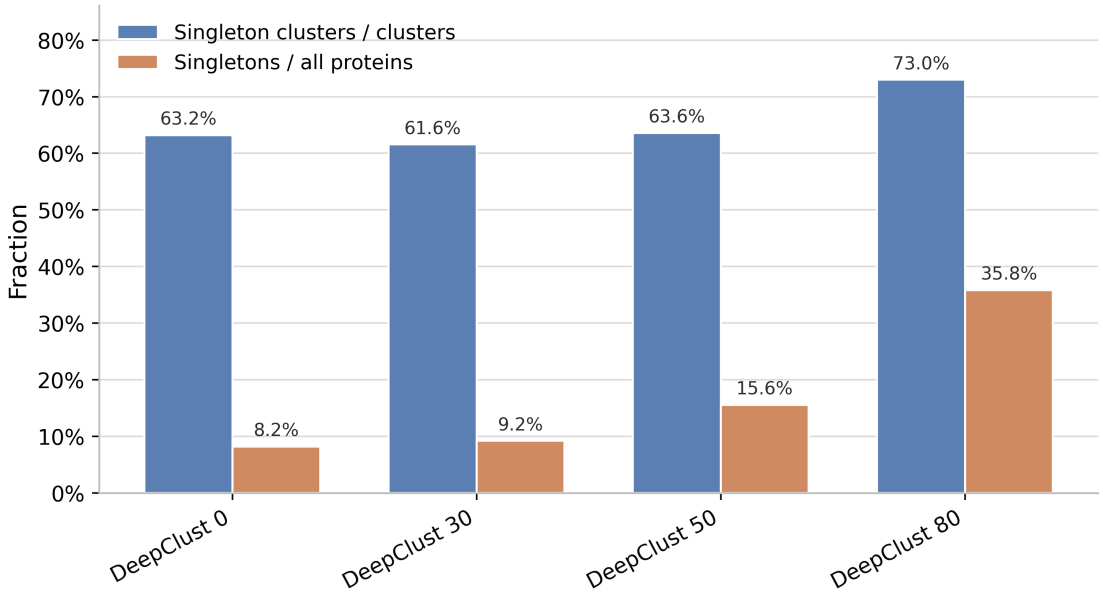


Figure 5: Singleton rates across DeepClust partitions, reported both as the fraction of clusters with a single protein and as the fraction of proteins assigned to singleton clusters.

Binning clusters by size further revealed that while DeepClust 80 assigned most proteins to small clusters of 2–10 proteins (40% of the initial 34.6M set), DeepClust 50 partitioned the largest fraction of proteins into clusters of 11–100 proteins (35%), and DeepClust 30 and 0 shifted the dominant protein fraction toward even larger clusters of 101–500 proteins (30% and 32%, respectively) with a substantial increase in proteins assigned to clusters with more than 500 members (Supplementary Fig. S4). This trend shows that clustering at lower stringencies does not only reduce the number of representatives, but progressively moves from small local clusters toward larger families capturing broader homology relationships. Accordingly, the distributions of cluster sizes across identity thresholds maintain a similar pattern for small cluster sizes, but progressively diverge in the tail, with lower stringency partitions showing a slower decrease and therefore substantially larger clusters (Supplementary Fig. S5). The maximum cluster size decreased from 14,309 proteins for DeepClust 0 to 7,331, 5,712, and 813 proteins for DeepClust 30, 50, and 80, respectively, illustrating how increasing stringency limits the formation of large broadly conserved families.

Based on these results, the no-threshold DeepClust 0 partition was retained as the primary clustering for downstream analyses, as it provided the strongest reduction in dataset size while limiting the proportion of proteins left as singletons.

3.2 Criterion-based validation and cluster refinement

The quality of the obtained clusters was assessed based on ultra-sensitive DIAMOND BLASTp searches between cluster members and representatives, as well as among representatives themselves. Assignment-based, intra- and inter-cluster metrics were computed as defined in Section 2.2.2.

In the initial DeepClust 0 partition, 30,129,558 non-representative members were evaluated,

of which 27.8M had at least one BLASTp hit to a representative. Among these proteins with hits, only 16.5M were assigned to their best hit representative (Fig. 6), corresponding to an accuracy of 59.2% among covered members and 54.8% overall (Fig. 7). The remaining proteins either had no BLASTp hit with their assigned representative or had a stronger match with another one.

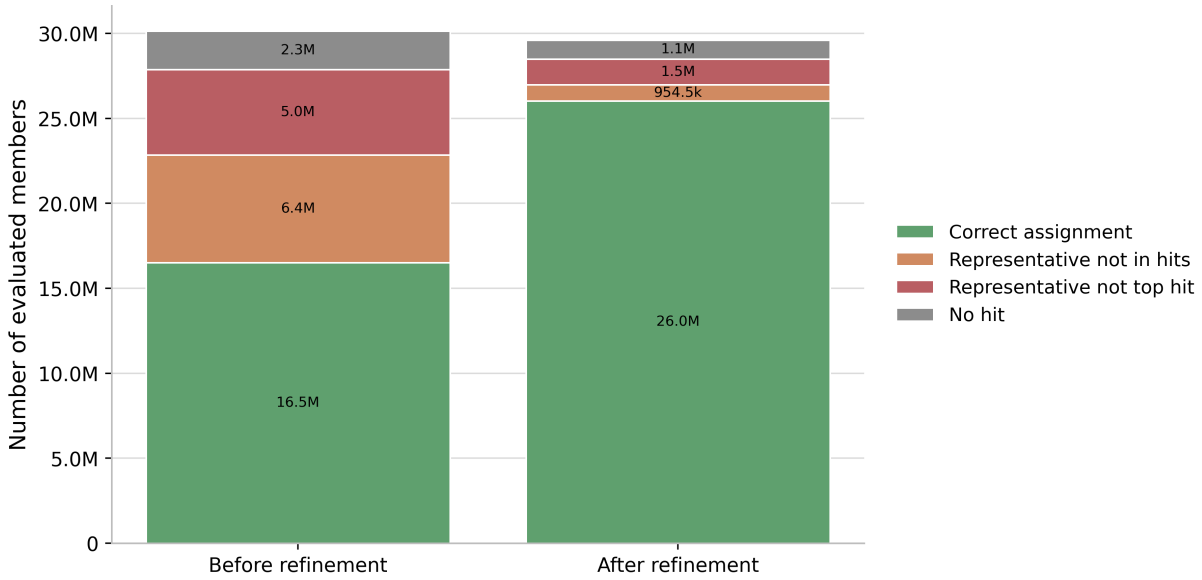


Figure 6: Member assignment outcomes before and after refinement. Outcomes distinguish correct assignments, members whose assigned representative was not recovered among BLASTp hits, members whose assigned representative was recovered but was not the best hit, and members with no representative BLASTp hit.

To correct these inconsistencies, the clustering was refined using DIAMOND *recluster* and *reassign* workflows (see Section 2.2.3). These corrections increased the total number of representatives from 4.5M to 5M, as a result of the splitting of clusters with problematic internal cohesion based on the clustering criterion. Accordingly, refinement shifted the cluster size distribution from being dominated by clusters of size 101–500 to a more balanced pattern, with roughly 30% of proteins falling into each of the 101–500 and 11–100 size bins. This substantially improved assignment consistency: 28.5M out of the total 29,604,425 evaluated members had at least one representative hit, and 26M were now correctly assigned (Fig. 6). Accuracy among covered members, thus, increased from 59.2% to 91.3%, with overall accuracy increasing to 88% (Fig. 7).

In parallel, representative-level matches were explored using an all-versus-all ultra-sensitive BLASTp search of the 4.5 million initial representatives, as well as the refined 5M set, to assess the extent of residual homology between clusters. In the initial DeepClust 0 partition, only 6.5% of representatives had at least one non-self BLASTp hit, corresponding to 11.7% of non-singleton representatives and 3.4% of singleton representatives (Supplementary Fig. S6). This fraction almost doubled to 12.4% after reclustering, reflecting a trade-off between intra-cluster cohesion and inter-cluster separation. By splitting broad heterogeneous clusters into finer groups with higher internal consistency, reclustering creates additional representatives that may still be homologous to one another, thereby increasing the proportion of representatives with detectable non-self hits. Notably, the reassignment workflow did not change the total number of matched representatives because it preserved the refined representative set, but it redistributed matches from singleton toward non-singleton representatives, consistent with proteins being reassigned to better fitting clusters.

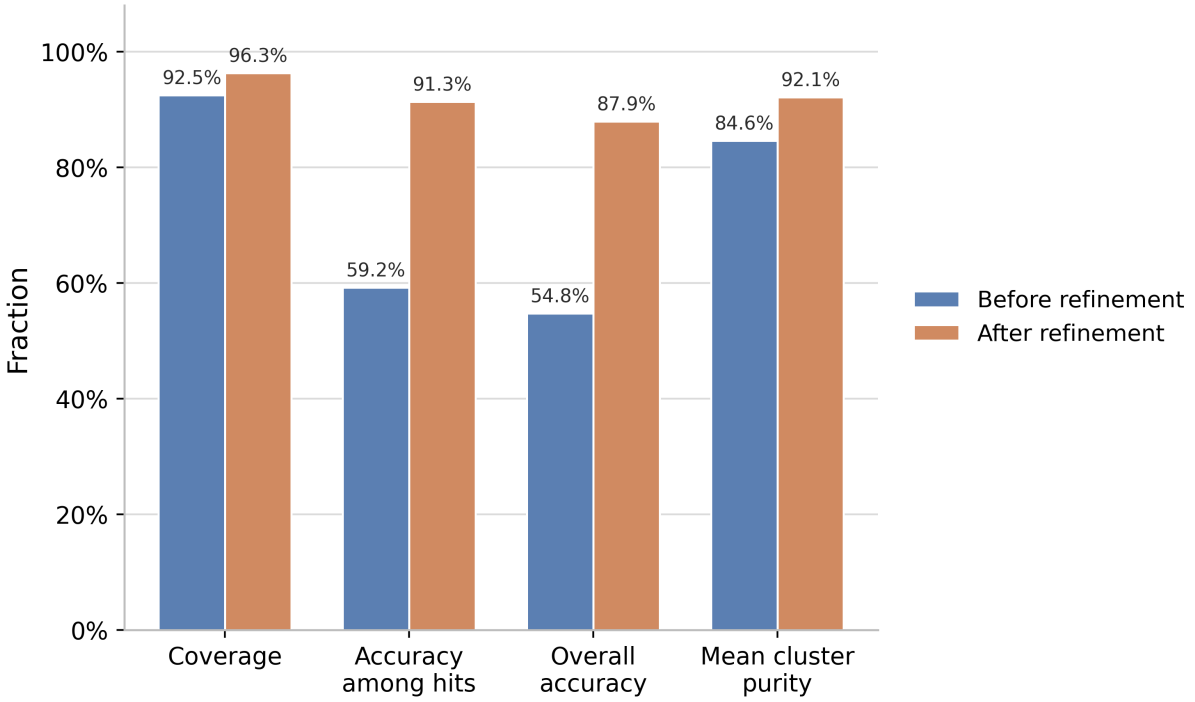


Figure 7: Criterion-based assignment quality before and after refinement. Refinement increased the fraction of members with at least one representative BLASTp hit, the accuracy among covered members, the overall assignment accuracy, and the mean cluster purity.

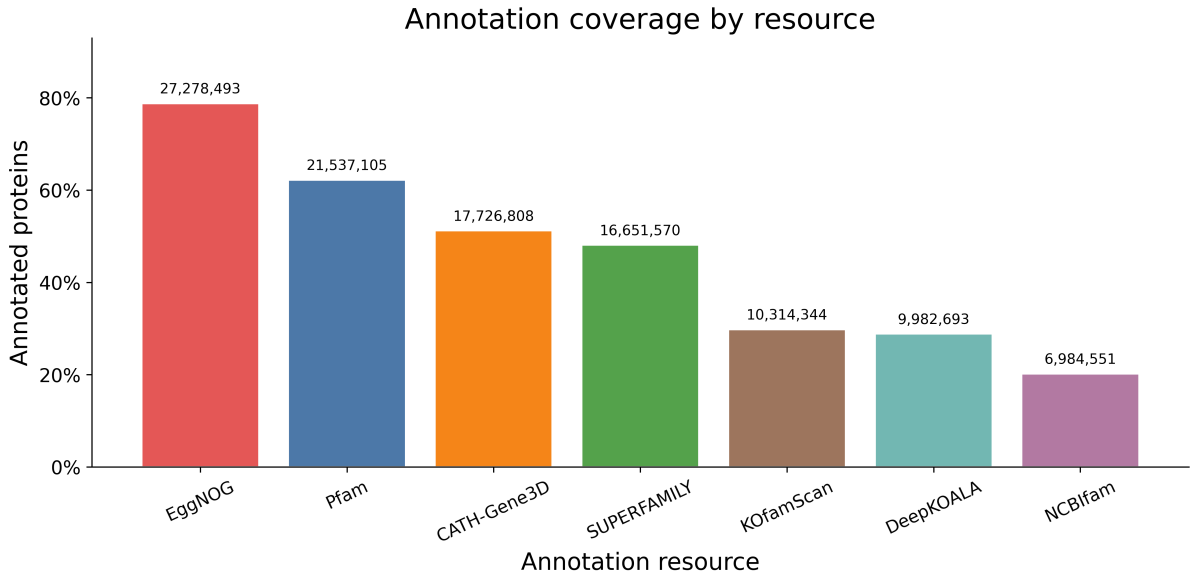
Further investigation showed that residual representative-level matches were concentrated in small cluster-size bins (Supplementary Fig. S7). This suggests that part of the residual representative redundancy could be reduced by merging small, closely related clusters after reclustering. Such a strategy will decrease representative-level matches and the total number of clusters while limiting the risk of reintroducing heterogeneity into large clusters. In contrast, merging larger clusters would be more likely to compromise the internal cohesion gained through reclustering.

Overall, criterion-based validation showed that the raw DeepClust 0 partition presented some assignment inconsistencies expected for large-scale cascaded clustering. The refinement steps successfully corrected most of these inconsistencies, strongly improving member-to-representative assignment accuracy and cluster purity while retaining a compact representative set for downstream analyses.

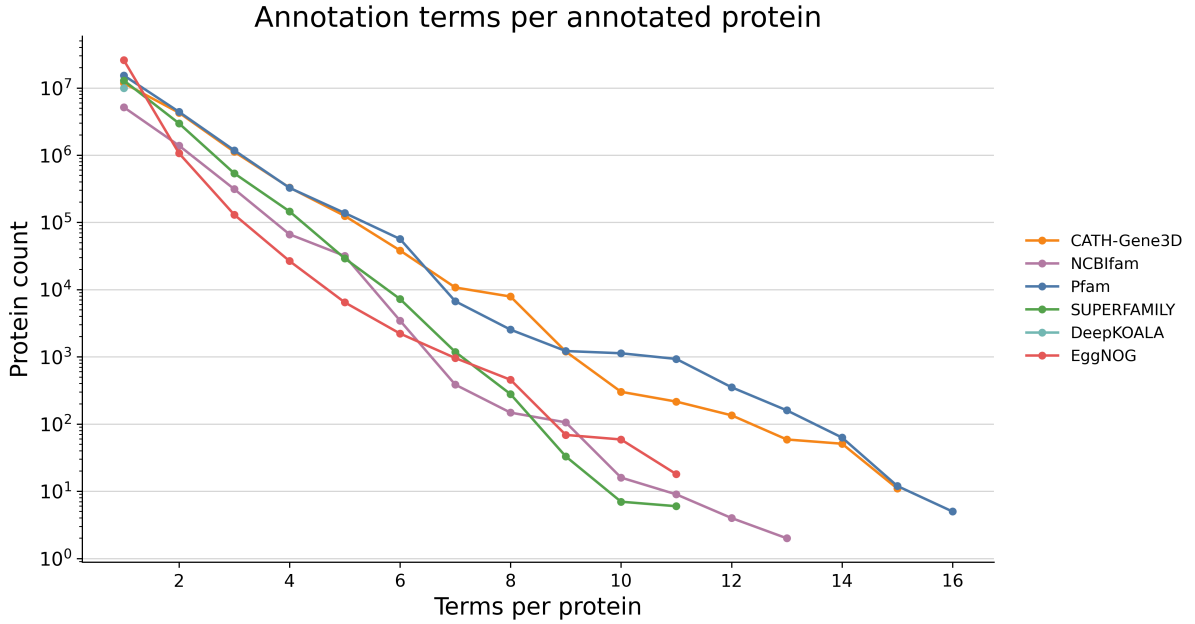
3.3 Multi-modal annotation across representatives

The full set of representative proteins across PanGBank pangenomes was annotated with all tools in our workflow (see Section 2.3), providing distinct views on protein domains, orthologous groups, functional modules, and structural superfamilies. EggNOG achieved the highest coverage, annotating close to 80% of all 34.6M representatives. On the other hand, NCBIfam had the lowest coverage, only annotating around 20% of all sequences (Fig. 8a), which is explained by its reliance on highly curated function-specific protein family models with strict confidence thresholds. Furthermore, most annotated proteins received one annotation per resource, yet the maximum number of annotations per protein ranged from 11 for SUPERFAMILY and EggNOG to 16 for Pfam (Fig. 8b), reflecting the presence of multi-domain proteins with multiple structural

or functional units. Notably, while DeepKOALA was configured to only keep the best hit per protein, KOfamScan reached a maximum of 35 annotations per protein.



(a) Coverage of each annotation resource. Labels indicate the number of annotated proteins.



(b) Distribution of the number of annotation terms assigned per annotated protein.

Figure 8: Coverage and annotation multiplicity across annotation resources.

Up to 6.6M proteins, corresponding to around 20% of the full representatives dataset, were left unannotated by any tool. Thus, besides the 8% of singletons (Fig. 5), another 12% of proteins display alignment-based homology relationships but lack annotation across resources. On the other hand, a major 17.5% of all sequences were annotated by four tools, which most commonly corresponded to Pfam, EggNOG, CATH, and SUPERFAMILY (Fig. 9). An additional 11M sequences (32%) were annotated by five or more tools.

Furthermore, pairwise overlaps between the annotations provided by the different tools were

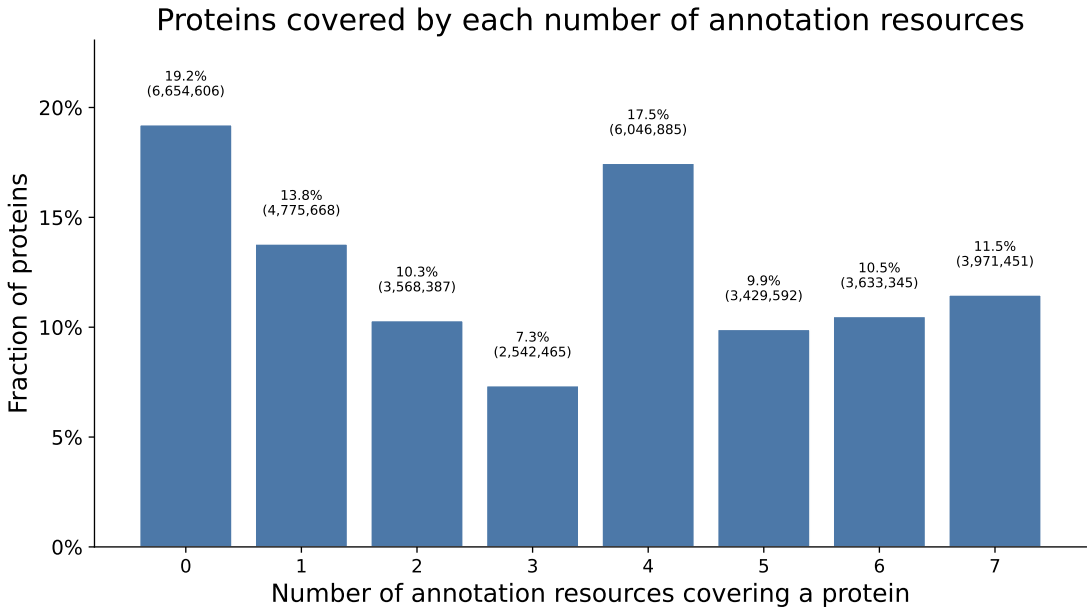


Figure 9: Distribution of proteins according to the number of annotation resources providing at least one annotation. Proteins in the zero-tool category were not covered by any considered resource, whereas higher values indicate broader multi-resource annotation support.

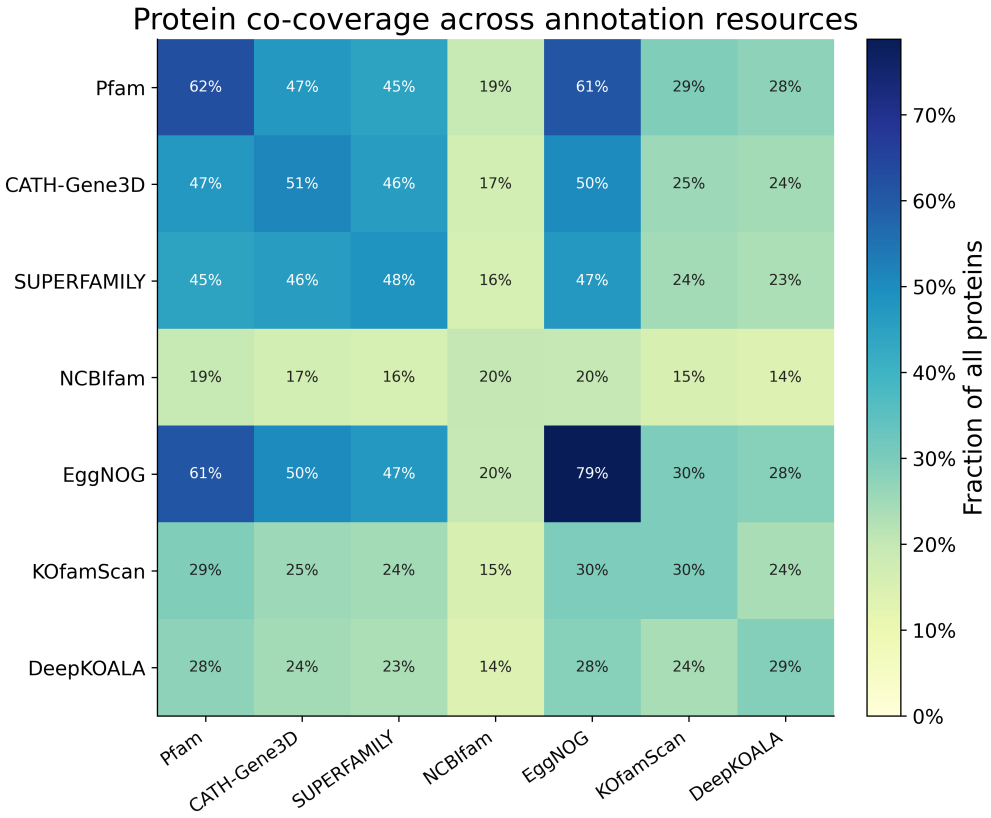
examined in terms of co-coverage and Jaccard similarity (Fig. 10). This revealed that almost all proteins covered by any tool were also covered by EggNOG (Fig. 10a). Considered together, co-coverage and Jaccard similarity suggest that the annotation resources do not primarily target distinct subsets of proteins, but they rather largely cover a common core of proteins, with some regions receiving annotations from multiple complementary resources and others only characterized by the most inclusive tools, particularly EggNOG. Notably, similar coverage did not necessarily imply redundant annotation content. Although CATH-Gene3D and SUPERFAMILY, as well as DeepKOALA and KOfamScan, target related annotation spaces and achieved comparable coverage, their pairwise Jaccard similarities showed only partial overlap, indicating that these tools annotate partly distinct proteins or assign different terms.

Overall, these results show that large-scale annotation of the PanGBank representative protein space yields broad but incomplete coverage, with each resource contributing a distinct layer of information, supporting the construction of a multi-modal annotation corpus.

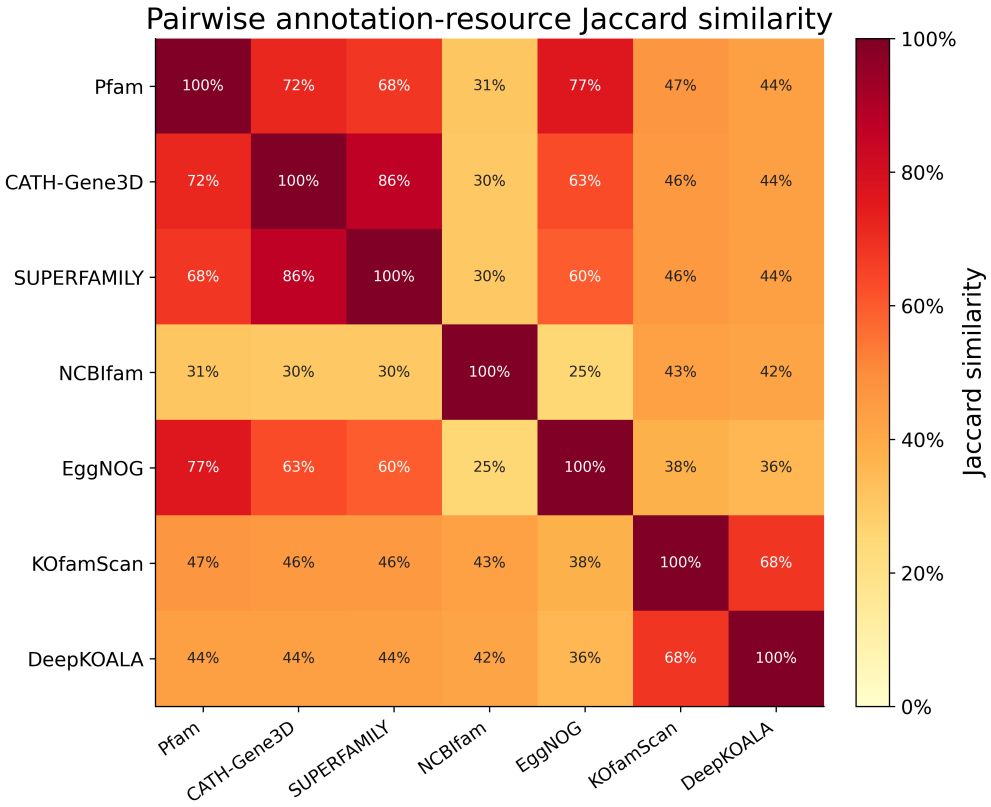
3.4 Functional consistency of alignment-based clusters

Subsequently, the functional coherence of DeepClust clusters was examined by computing the annotation agreement between the members of each cluster and their corresponding representative across all annotation tools. For each clustering level, exact-match rates were computed by comparing the complete set of annotations assigned to each member with those of its representative. In parallel, Jaccard similarity was used to account for partial agreement when proteins carried multiple annotations (see Section 2.2.2).

Annotation consistency increased with clustering stringency across all resources (Fig. 11). Notably, cluster refinement also corresponded to an increase in consistency more significant than raising the clustering identity threshold by 30%. In the no-threshold DeepClust partition, exact-match rates ranged from 56% for Pfam to 81% for NCBIFAN. However, this is, in part, due to



(a) Pairwise protein co-coverage between annotation resources. Diagonal values represent individual resource coverage.



(b) Jaccard similarity between the sets of proteins covered by each resource.

Figure 10: Overlap and complementarity between annotation resources. Co-coverage is calculated relative to the complete protein collection, whereas Jaccard similarity normalizes pairwise overlap by the union of the covered protein sets.

the large number of missing annotation for NCBIfam, inflating its apparent consistency value. A corrected measure was thus examined after restricting the comparison to member-representative pairs where both proteins were annotated by the considered resource. This led to a slight decrease in consistency scores across tools, with Pfam remaining the lowest and DeepKOALA becoming the highest (Supplementary Fig. S9b).

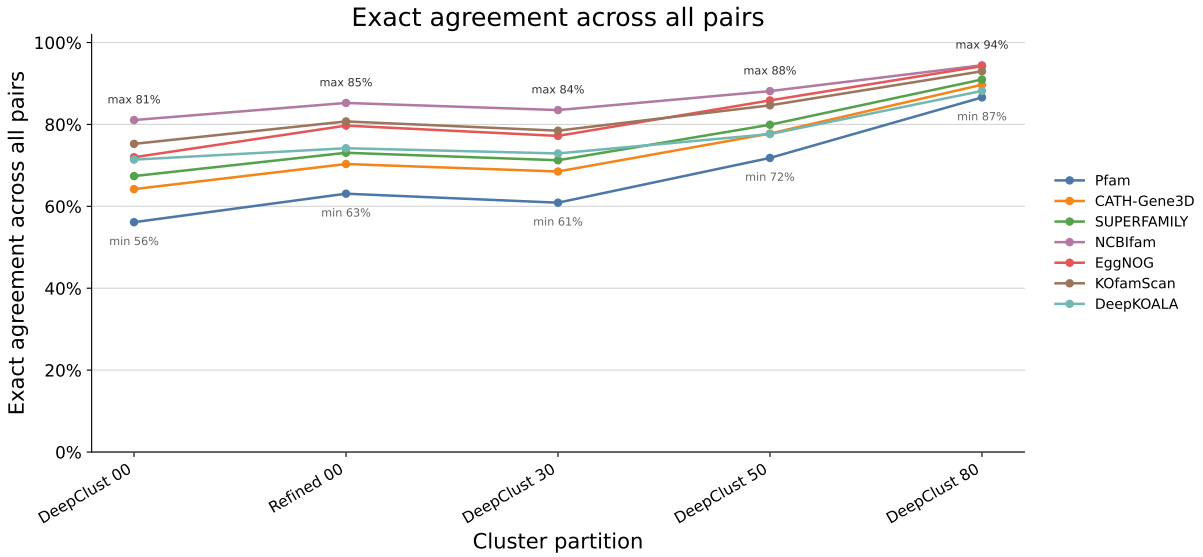


Figure 11: Exact annotation agreement between cluster representatives and members across clustering partitions. Agreement was evaluated independently for each annotation resource across all representative–member pairs.

Alternatively, the Jaccard agreement scores revealed significant partial matches for proteins with multiple annotations per tool, especially for lower stringency partitions. Jaccard consistency scores were notably higher, ranging between 74% for DeepKOALA and 88% for NCBIfam even for the least stringent DeepClust 0 after refinement (Supplementary Fig. S9a).

Overall, DeepClust alignment-based clusters are revealed to preserve substantial functional consistency, even with no identity cut-off. In addition, cluster refinement was revealed to increase consistency even more significantly than raising the sequence identity stringency. These findings also confirm the 80% identity threshold as significant enough to transfer annotations from representatives to members, suggesting that future annotations could be executed on this reduced subset of representatives.

3.5 Embeddings provide complementary insights

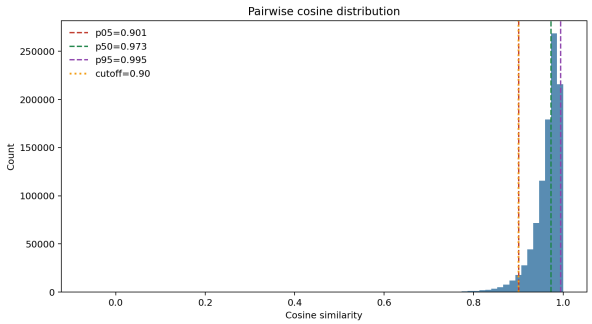
Protein embeddings were computed to provide an AI-derived representation of PanGBank representative sequences complementary to alignment-based clusters and curated annotations. To this end, ESM embeddings were primarily computed for the final representatives obtained with DeepClust 0 (see Section 2.4). The smaller ESM 150M model was further applied to a substantially larger dataset covering approximately 17 million sequences, nearly half of the total representatives across PanGBank pangenomes.

Whereas DeepClust relies on local sequence similarity and explicit clustering criteria, ESM represents each protein as a high-dimensional embedding matrix derived from patterns learned across hundreds of millions of protein sequences. As a result, embedding similarity can cap-

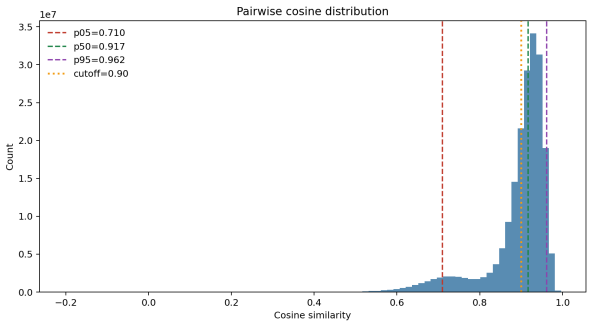
ture structural and evolutionary relationships that are undetectable by traditional approaches. Therefore, ESM embedding space was leveraged to evaluate whether DeepClust clusters remain coherent beyond their original clustering criterion, to explore the spatial organization of annotated and poorly annotated proteins, and to construct broader embedding-based superclusters.

3.5.1 Consistency of DeepClust clusters in ESM space

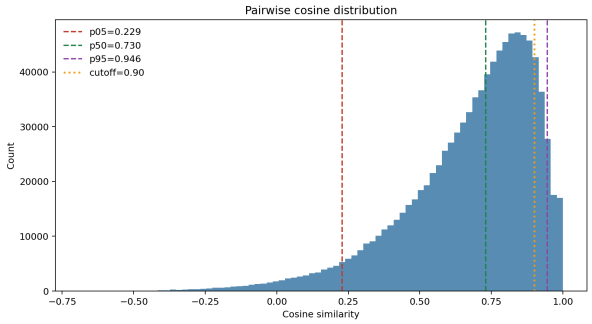
The consistency of DeepClust clusters in ESM space was assessed by comparing the embeddings of cluster members with those of their corresponding representatives, as detailed in section 2.4. Our initial computations revealed surprisingly high cosine similarity, even between DeepClust cluster representatives, which should correspond to biologically unrelated proteins (Fig. 12b). Further investigation revealed this inflated similarity, even between random or unrelated proteins, to be due to a strong anisotropy in the raw ESM embedding space, where embeddings occupy a narrow cone of the vector space. After considering several normalization, pooling, and transformation strategies, we identified the centering transformation as an effective solution to this issue, yielding a perfectly normal distribution between cluster representatives, as shown in Fig. 12d.



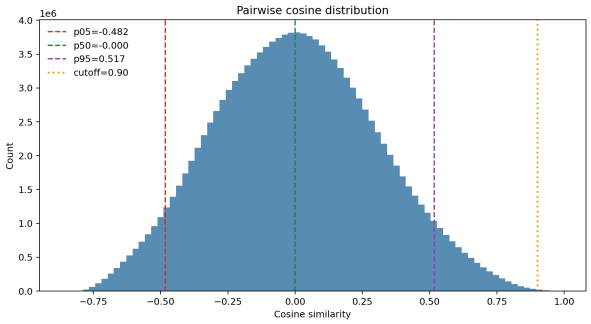
(a) Member-to-representative similarities before centering.



(b) Representative-to-representative similarities before centering.



(c) Member-to-representative similarities after centering.



(d) Representative-to-representative similarities after centering.

Figure 12: Distributions of cosine similarities between cluster members and their corresponding representatives and among representatives, before and after applying the centering transformation to ESM embeddings. Raw embeddings show seemingly high cosine similarity even between distinct representatives, reflecting embeddings anisotropy. After centering and L2 normalization, representative-to-representative similarities are centered near zero, while member-to-representative similarities remain high.

While centered representative-to-representative cosine similarities followed a normal distribution with a mean of 0.005 and a median of 0, member-to-representative similarities remained

substantially higher, with a mean of 0.677 and a median of 0.730 (Fig. 12c). The centered embedding space therefore supports the biological consistency of DeepClust clusters. In a direct observed-versus-null comparison using 300,000 sampled pairs, 79% of observed member-to-representative pairs exceeded the 95th percentile of the null distribution, and 55.0% exceeded the 99th percentile, thus confirming that DeepClust members remain significantly closer to their assigned representatives in ESM space than expected from background representative similarity (Supplementary Fig. S10).

3.5.2 Organization of annotations in ESM space

The ESM embedding space was then used to examine the organization of functional annotations. This was restricted to the final DeepClust 0 representatives, which were projected into a two-dimensional landscape with incremental PCA and colored according to InterProScan annotation coverage (see Section 2.4). Among the full set of the 4.5M DeepClust 0 representatives, 1,007,987 had at least one InterProScan annotation, corresponding to 22.4% of all representatives and 60% of non-singleton representatives. Consistent with the reported findings across the full set of 34.6M representatives (Section 3.3), Pfam maintained the broadest coverage, annotating 767k representatives (17%), followed by CATH-Gene3D at 15.5%, SUPERFAMILY at 14%, and NCBIfam with only 122k representatives (2.7%).

The projection revealed a gradual spatial transition from densely annotated regions to regions with progressively fewer annotations (Fig. 13). This organization suggests that annotation absence is not randomly distributed across the embedding space. Instead, poorly annotated proteins occupy coherent regions, consistent with the idea that part of the microbial protein "dark matter" corresponds to structured but under-characterized areas of protein sequence space. The same pattern persists when each annotation tool is considered separately (Supplementary Fig. S11).

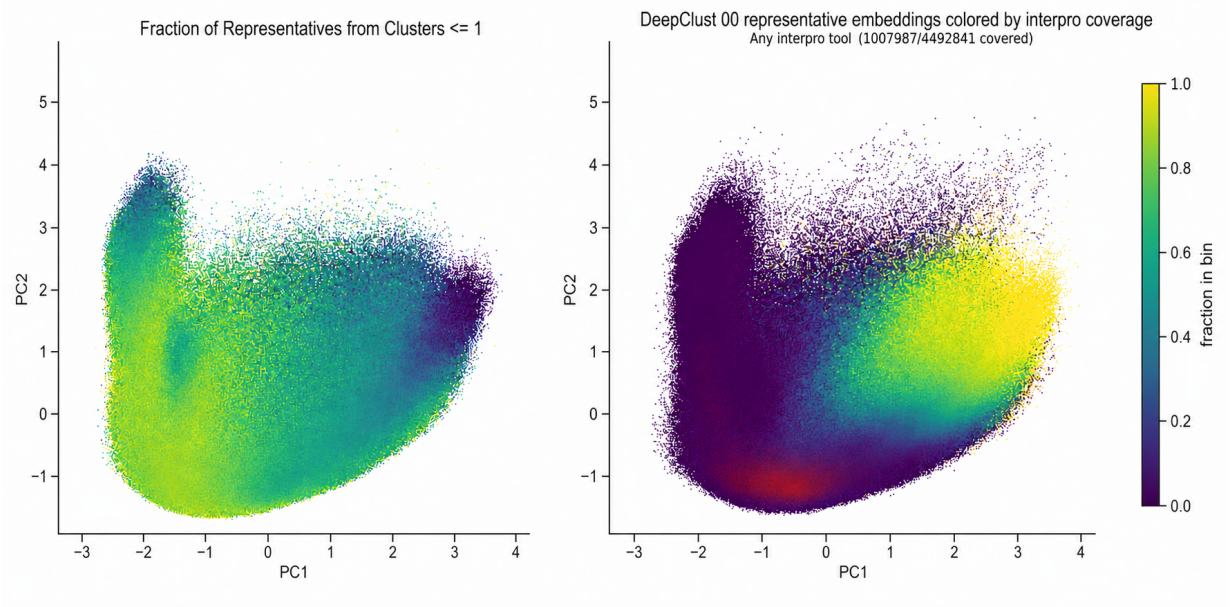


Figure 13: ESM embedding-space projections of annotation coverage and cluster size. (Left) Local fraction of DeepClust singleton representatives; (Right) InterPro annotations coverage.

To further interpret these poorly annotated regions, shuffled protein sequences derived from the ASTRAL40 dataset were included as negative controls, following the approach suggested

by Prabakaran and Bromberg [64]. These shuffled sequences preserve amino acid composition while disrupting biological sequence organization. Their overlap with part of the poorly annotated embedding landscape suggests that some dark regions may correspond to proteins with weakly informative or composition-driven representations. However, the existence of coherent poorly annotated regions beyond the random controls also suggests that embeddings may help prioritize candidate protein families that lack current annotations but retain non-random sequence organization.

Notably, as highlighted in the leftside panel of Fig. 13, the density of singletons in ESM space followed a directly opposite pattern, which also persisted for small clusters (<= 5).

3.5.3 ESM-based superclusters partitioning

Finally, ESM embeddings were used to construct a higher-level partition of representatives (see Section 2.4). After centering and L2 normalization, an approximate nearest-neighbor graph was built from representative embeddings and clustered into embedding-based superclusters. This produced 72 superclusters across the 4.5M DeepClust representatives. The obtained superclusters were highly uneven in size: the largest supercluster contained 412,881 representatives, corresponding to 9.2% of the dataset, while the ten largest superclusters together accounted for 60% of all representatives (Supplementary Fig. S12).

The two-dimensional projection of sampled representatives showed that the superclusters mostly occupy coherent regions of the embedding landscape (Supplementary Fig. S13a), indicating that the graph partition captures large-scale structures in the centered ESM space rather than producing arbitrary communities. While most cluster sit close to each other forming a prolonged contiguous landscape, the density map reveals distinct regions of high density, which support the clustering algorithm partitions (Supplementary Fig. S13b).

Similarity diagnostics further support the coherence of the superclusters partition. Random pairs sampled within the same supercluster had a mean cosine similarity of 0.54 and a median of 0.57, whereas random pairs sampled between different superclusters had a mean similarity of 0.07 and a median of 0.08 (Fig. 14). Moreover, boundary pairs connected by the nearest-neighbor graph showed higher similarity, with a mean of 0.79, reflecting local connections between adjacent superclusters, consistent with the obtained UMAP pattern. Overall, ESM-based superclusters appear to capture broad regions of high internal similarity while preserving local continuity between neighboring regions of the embedding space.

Annotation-based Validation The functional content of superclusters was examined by measuring InterProScan enrichment and coverage across them. Enrichment patterns showed that several superclusters were associated with distinct annotation profiles, suggesting that the embedding partition captures biologically meaningful organization beyond purely alignment-based clustering, as highlighted in Supplementary Fig. S14 and its associated Supplementary Table S1. Indeed, as ESM-based superclustering was conducted at the level of the representatives obtained with the least stringent DeepClust 0 partition, sequence similarity hits are almost completely missing among representatives, with only 6% of them showing any BLASTp ultra-sensitive hits (Supplementary Fig. S6), most of which corresponded to very small clusters (Supplementary Fig. S7). This confirms that the performed ESM-based clustering was not done on the basis of

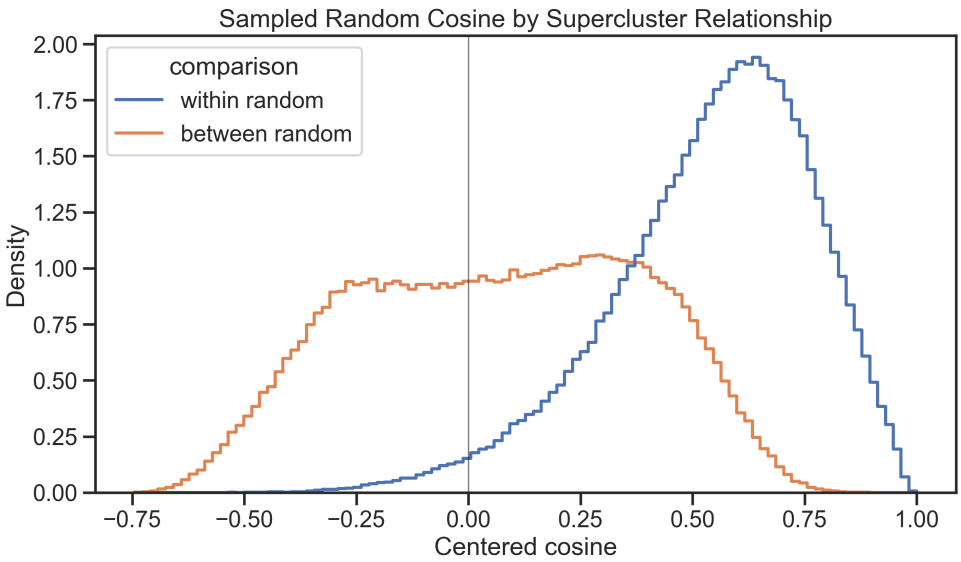


Figure 14: Cosine similarity diagnostics for ESM-based superclusters. Within-supercluster random pairs are substantially more similar than between-supercluster random pairs.

direct sequence similarity but other high-dimensional insights learned through ESM embeddings. Furthermore, this makes the successful recovery of functional enrichment patterns among these superclusters particularly impressive, providing direct ground for annotation transfer on the basis of embedding similarity even where alignment-based hits are completely absent.

Structural Validation Besides functional enrichment, the extent to which the members of each supercluster displayed structural similarity was investigated as described in Section 2.4. Same-supercluster protein pairs were found to be more frequently observed at higher structural similarity scores than between-supercluster pairs, as shown in Figure 15. This pattern is significant under label permutation testing (same-supercluster discrimination AUC = 0.673, permutation null mean $p \approx 0.002$), which assesses significance relative to a null distribution generated by randomly reshuffling the supercluster labels.

Moreover, structural coherence increased substantially after restricting the analysis to structures with high-confidence predictions. At median pLDDT $\geq$ 80, same-supercluster AUC increased to 0.765 and structural nearest-neighbour recovery reached 48.8%, compared with a prevalence baseline of 2.8%. This pattern was especially clear among the strongest structural matches: within the top 0.1% of Foldseek-ranked pairs, 76% linked proteins from the same ESM supercluster, corresponding to a 27-fold enrichment over baseline. These results suggest that prediction uncertainty attenuates the structural signal, while the highest-confidence and highest-similarity pairs provide strong support for structural coherence within the ESM-derived superclusters (Supplementary Fig. S15). Overall, structural similarity signals were modest in absolute similarity (TM-score), as expected for superclusters built from already broad DeepClust representatives, but the direction is consistent: within-supercluster pairs are more structurally similar than between-supercluster pairs, and the pattern improves for better predictions.

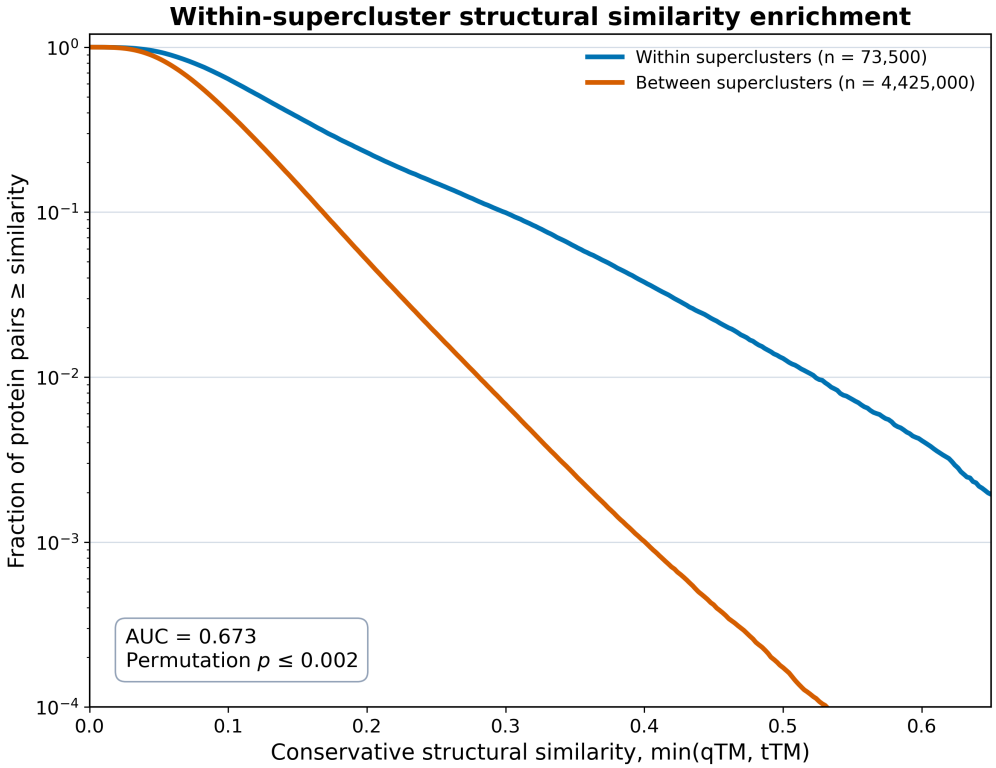


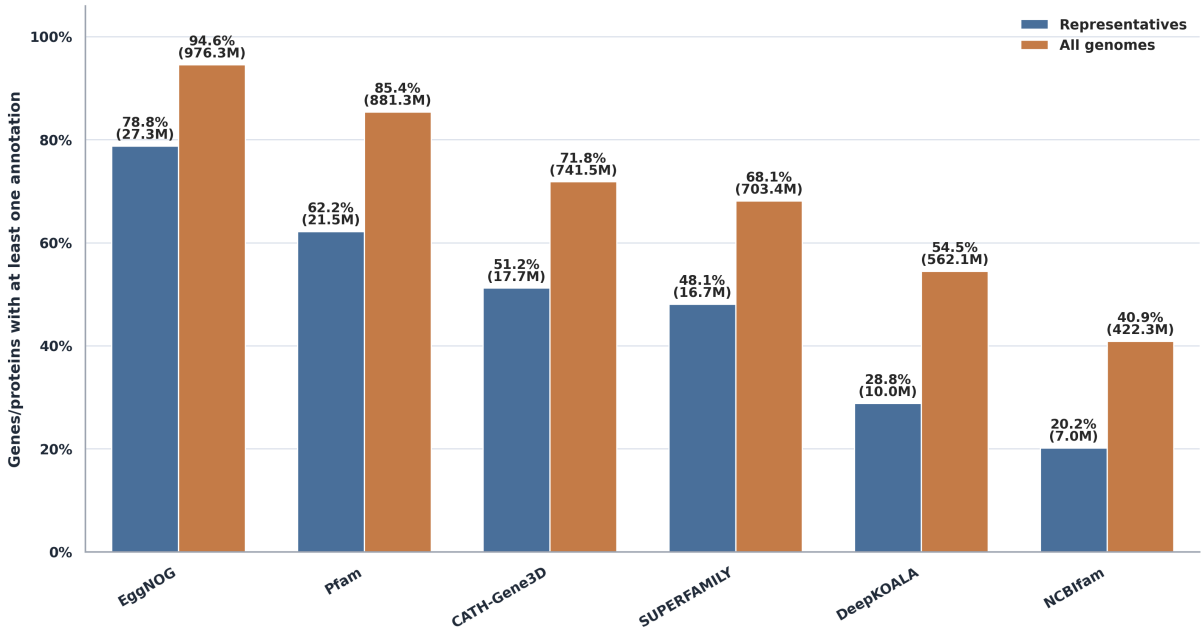
Figure 15: Structural similarity enrichment within ESM-derived superclusters. Same-supercluster and between-supercluster pairs were compared using conservative Foldseek similarity scores, with enrichment assessed against a label-permutation null distribution.

3.6 Construction of a large-scale annotation corpus for LLM training

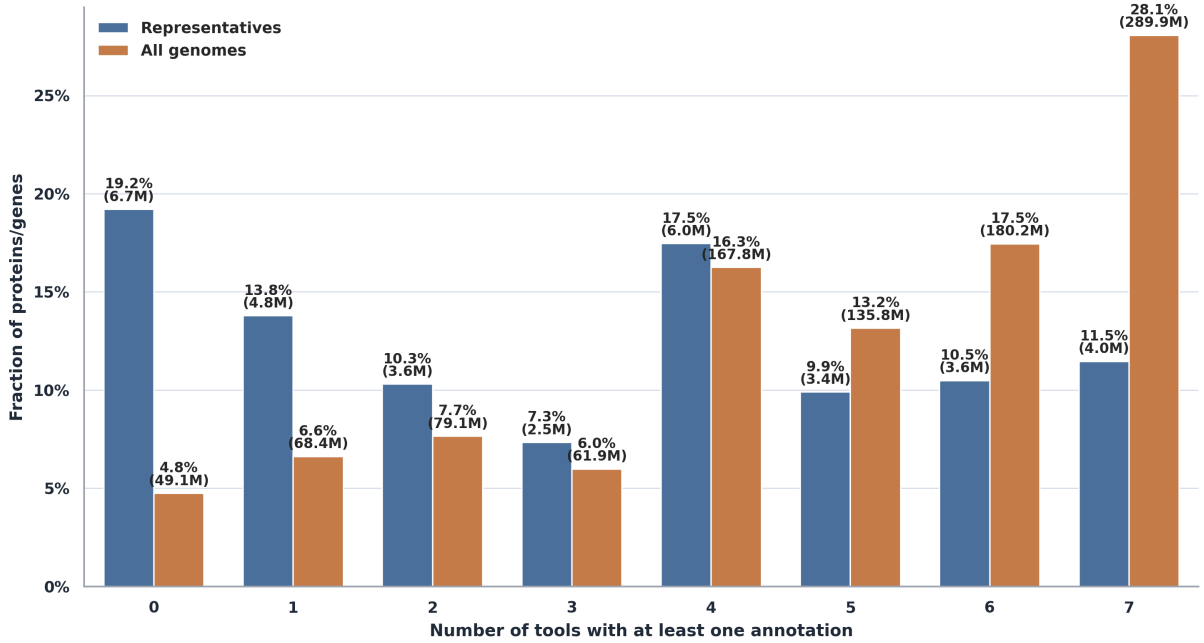
The different annotation vocabularies were regarded as parallel layers and tokenized as presented in Fig. 3, following propagation to all genomes of each pangenome, vocabulary setup, and analysis. Vocabulary optimization was also explored prior to tokenization.

Annotations Propagation After obtaining the annotations across the representative protein families of all PanGBank pangenomes, they were propagated back to all constituent genomes in order to generate a substantially larger annotation corpus for the downstream LLM training stage (see Section 2.5). Accordingly, the protein space underwent a 30-fold expansion, from the 34.6 million initial representatives to 1.032 billion proteins (Table 1). Moreover, this propagation led to a significant increase in the total coverage across all annotation tools (Fig. 16a), since most non-annotated representatives corresponded to singletons or very small families which are much less represented in the full protein space compared to the larger annotated families. Notably, the propagation step also allowed the extraction of the modules and RGPs identified by panModule and panRGP and stored in each PanGBank pangenome, providing additional contextual information that may be exploited in downstream analyses.

Across the full protein space, the proportion of "dark matter", or proteins with no annotation across sources, decreased from 19% at the representative level to only 4.8%. Similarly, the proportions of proteins covered by 1–4 annotation resources decreased alongside a significant increase in the proportions of proteins covered by 5–7 tools. In particular, proteins covered by all seven tools represented up to 28% of the full propagated protein set (Fig. 16b).



(a) Annotation coverage at the level of PanGBank representative proteins compared to the complete protein set across all genomes.



(b) Number of annotation tools covering each representative protein in PanGBank compared to the complete protein set across all genomes.

Figure 16: Comparison of annotation coverage at the representative level and all-genome level. The two views summarize differences in overall annotation coverage and in the number of annotation sources contributing to each protein before and after annotation propagation to all genomes.

This propagation also revealed the great discrepancies in pangenome sizes in PanGBank, with 72% of pangenomes representing fewer than 50 genomes, and a few representing up to 2000 genomes. These size differences will lead to substantial differences in the absolute contribution of each pangenome to the final corpus (Supplementary Fig. S16). Yet, global and per-tool annotation coverages remain comparatively stable across pangenome sizes, with only weak correlations between the number of genomes per pangenome and either the fraction of annotated genes (Pearson $r = 0.098$; Supplementary Fig. S16b) or the mean number of tools per gene (Pearson $r = 0.081$; Supplementary Fig. S16a). Thus, no notable increase in annotations was observed for larger pangenomes, and the same per-tool coverage ranges noted at the representative level were maintained across pangenome sizes.

Vocabulary Analysis The different annotation layers obtained across PanGBank representative proteins were analyzed as candidate vocabularies for downstream LLM training, where annotation terms are treated as discrete tokens and genomic contexts are encoded as ordered sequences of these tokens. To this end, the size, sparsity, redundancy, and frequency structure of each annotation vocabulary were examined.

Annotation resources differed strongly in vocabulary size (Fig. 17). EggNOG provided a disproportionately larger vocabulary of 163.5 thousand unique terms, even after retaining only root orthologous groups and discarding lower-level classifications. Pfam came second with 18.5k distinct terms and 30M total term mentions (1.28B after propagation to all genomes). DeepKOALA and NCBIfam produced comparably large vocabularies, with 18.2k and 17k distinct terms, respectively, despite NCBIfam having a substantially lower coverage. CATH-Gene3D and SUPERFAMILY provided the most compact vocabularies, with 5,170 and 1,860 terms, respectively, despite having 26M and 21M total mentions (1.1B and 949M after propagation to all genomes), respectively.

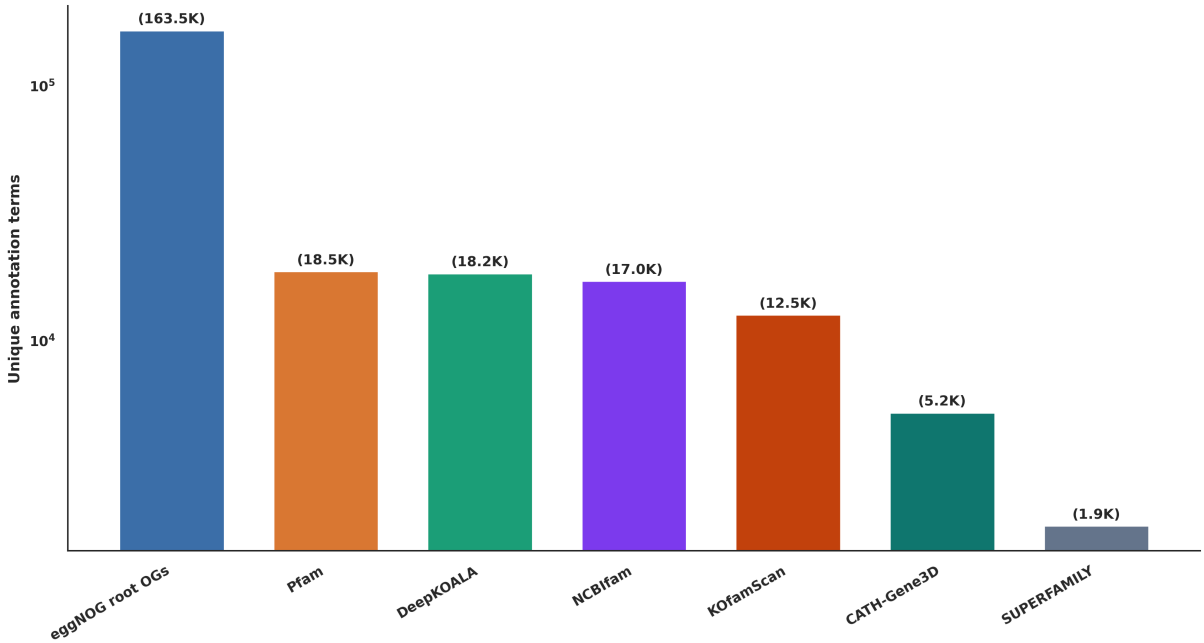


Figure 17: Vocabulary size across annotation tools for representative proteins. The number of distinct retained terms differs substantially between annotation sources, reflecting differences in annotation scope, granularity, and filtering strategy.

The annotation vocabularies also showed strong frequency imbalance, a property expected for biological token spaces where a limited number of common domains or structural classes coexist with many rare terms. This imbalance was quantified by high Gini coefficients (see Section 2.5), ranging from between 0.8 and 0.9 across all tools. While the top 100 terms accounted for close to 20% of the total annotation mentions of most tools, CATH and SUPERFAMILY showed an exceptional dominance of 54% and 62% of their total term mentions, respectively, by their top 100 terms (Fig. 18).

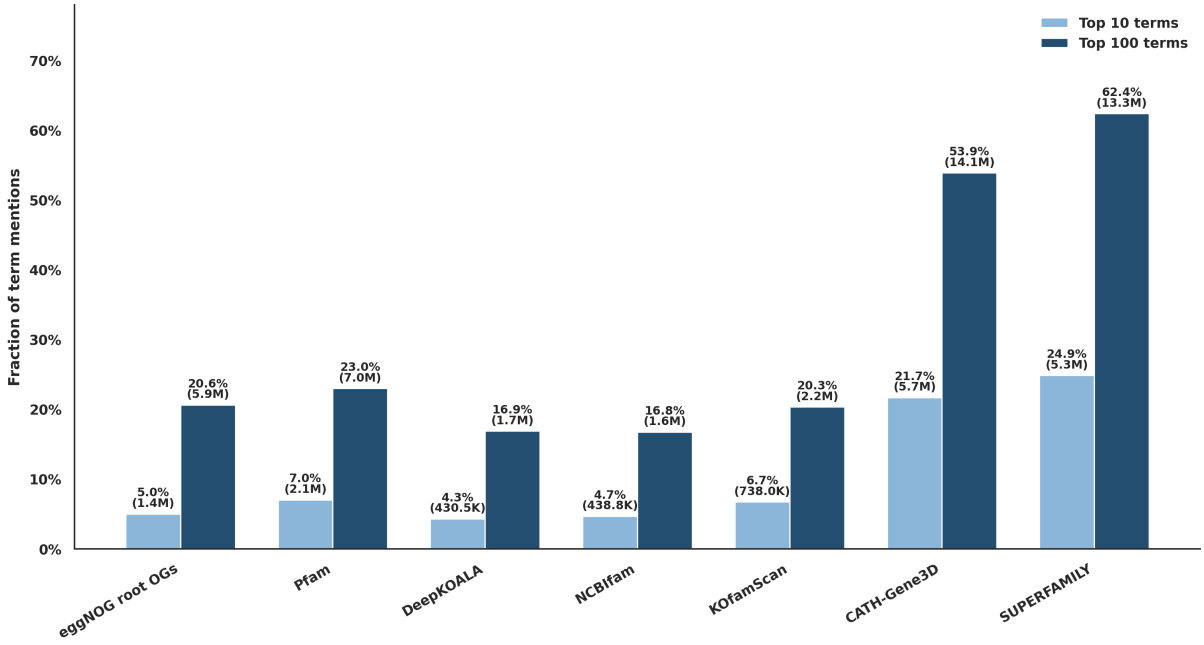


Figure 18: Coverage of representative proteins by the most frequent annotation terms. The fraction of proteins covered by the top 10 and top 100 terms highlights the extent to which each annotation vocabulary is dominated by highly recurrent terms.

Finally, the rank-frequency distributions were found to follow a strong long-tailed pattern across resources (Fig. 19). CATH and SUPERFAMILY displayed the most Zipf-like pattern with an alpha close to 1 (see Section 2.5), confirming their vocabularies resemblance to natural-language token distributions, whereby a small set of frequent tokens provides broad coverage while the remaining terms become increasingly rare. On the other hand, notable deviations were identified for the vocabularies of the other tools. Accordingly, the final vocabularies could be optimized to balance coverage and granularity. However, direct filtering or subsampling would be insufficient, as this would simplify training but risk removing biologically specific functions. Moreover, it is unclear how such modifications could be performed without disrupting the underlying distributional semantics of the vocabulary.

Vocabulary Optimization Clustering-based optimization of annotation vocabularies was explored through a sequence homology-informed aggregation of annotation terms per tool. First, to assess the reliability of this approach, the consistency between Pfam clans and the Pfam families associations based on the DeepClust 0 partition was examined. In this context, Pfam clans are consensus higher-level groupings of related Pfam families that share a common evolutionary origin but may be too divergent to be detected as a single family. Accordingly, only DeepClust 0 clusters with members corresponding to at least two Pfam families were retained, amounting to a

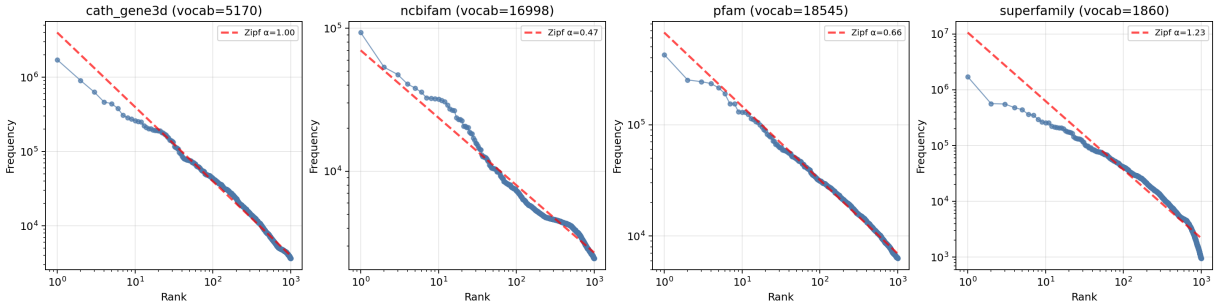
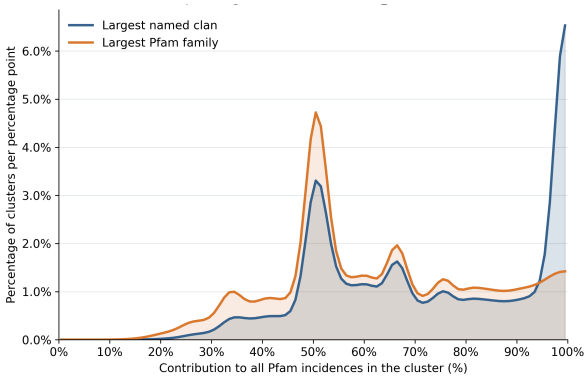


Figure 19: Rank-frequency structure of InterProScan annotation vocabularies. The long-tailed distributions indicate that annotation tokens follow a Zipf-like organization, with a small number of frequent terms and a large number of rare biological terms.

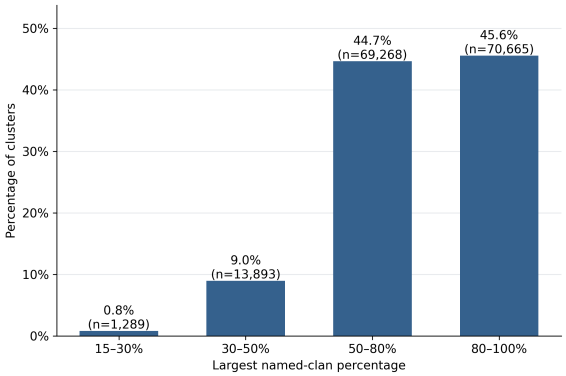
total of 155,115 clusters. Among these clusters, around 45% were found to have 50–80% of their Pfam families belong to one clan, with another 45% having at least 80% of their families within one clan (Fig. 20b). In addition, the distribution of maximal per-family and per-clan contributions within multi-Pfam DeepClust clusters revealed a significant peak near 100% at the clan-level and a smaller peak close to 50% at the family-level (Fig. 20a). Together, these results strongly suggest that DeepClust could be leveraged for a biologically informed refinement of Pfam vocabulary, and potentially for the other annotation vocabularies as well.

To this end, Pfam families were grouped using their co-occurrence inside DeepClust clusters. For each cluster, Pfams were considered associated if they were both present in at least 10% of the annotated proteins in that cluster. For each Pfam, its strongest matching Pfam association was retained as a representative, and these retained best-match links were then transitively merged into connected components, producing DeepClust-associated Pfam groups. Unlinked Pfams remained as single-family tokens. This strategy reduced the original 18.5k total Pfam terms to only 9.5k terms, notably similarly to the reduction at the clan-level, which corresponds to a total of 10.3k terms.

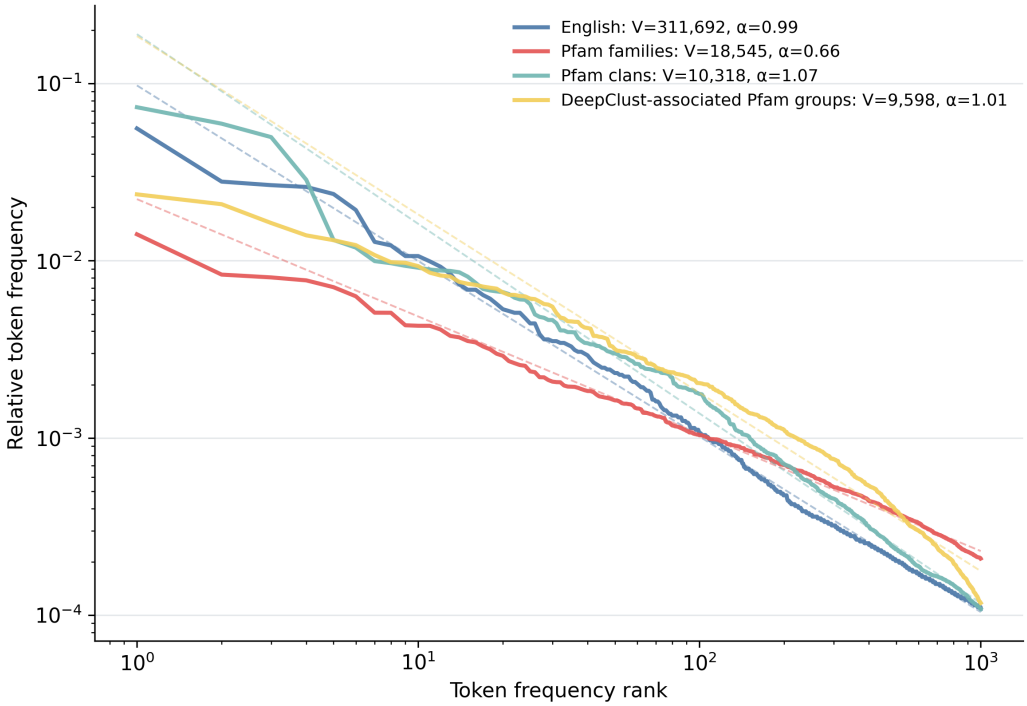
Finally, the quality of this vocabulary refinement was assessed based on its effect on the Zipf distribution. The DeepClust-based reduced vocabulary produced the closest behavior to the canonical Zipf law, with a fitted exponent of $\alpha = 1.006$, compared with $\alpha = 0.662$ for the original Pfam family vocabulary (Fig. 20c). The clan-based reduction also substantially improved the distribution, with $\alpha = 1.070$, although it was slightly farther from the ideal $\alpha \approx 1$. For reference, the English language frequency distribution and fitted exponent are also shown in the same plot. Overall, this suggests that DeepClust-derived grouping is consistent with the known Pfam clans while also capturing broader recurrent annotation combinations, which motivates the exploration of extending this refinement strategy to the other annotation vocabularies.



(a) Clan- and Pfam-family dominance within multi-Pfam DeepClust clusters.



(b) The proportion of multi-Pfam DeepClust clusters with one Pfam clan covering each displayed range of total Pfam terms percentage.



(c) Zipf comparison of raw Pfam families, clan-collapsed Pfam tokens, and DeepClust-associated Pfam groups.

Figure 20: DeepClust supports homology-driven grouping of Pfam annotations. Multi-Pfam DeepClust clusters show dominant clan-level structure. At the vocabulary level, DeepClust-associated Pfam grouping improves the rank-frequency profile relative to raw Pfam families: Zipf exponent increases from $\alpha = 0.662$ for raw Pfam families to $\alpha = 1.006$ after DeepClust grouping.

4 Discussion

This work established a scalable framework for characterizing and tokenizing the protein content of PanGBank. By combining alignment-based clustering, multi-modal annotations from complementary curated resources, and protein language-model embeddings, the developed workflows transformed 34.6 million proteins into a structured corpus suitable for downstream LLM training. The results also demonstrate that no individual representation fully captures protein diversity: sequence similarity, annotations, and embeddings provide related but complementary insights.

Clustering DeepClust proved to be exceptionally efficient in terms of computational runtime and protein space reduction. It managed to reduce the 34.6M total representatives from across PanGBank pangenomes to 4.5M representatives in less than 2 hours. In addition, the resulting clusters demonstrated strong quality across multiple validation strategies. Criterion-based validation using ultra-sensitive BLASTp confirmed adherence to the intended clustering criteria, annotation-based validation revealed high functional consistency within clusters, and embedding-based validation showed that cluster members were substantially more similar to their representatives compared to inter-cluster similarities in the embedding space. The high rate of singletons observed is expected for the clustering of large protein repertoires. DeepClust reported a 68% singletons rate in its clustering of 19B total sequences from across the tree of life at 30% sequence identity and 90% coverage [38]. In comparison, our DeepClust 30 partition yielded a 62% singletons rate for PanGBank 34.6M sequences at 80% coverage (Fig. 5).

Furthermore, cluster refinement, as highlighted in Fig. 1, was revealed to further improve the obtained cluster quality, leading to significant improvements in cluster purity and assignment error rate. However, this improvement came at the expense of a slight increase in the rate of inter-cluster representative matches, demonstrating a trade-off between inter- and intra-cluster qualities optimization. Still, most of the inter-cluster matches introduced after refinement were found to correspond to singletons and very small clusters, making them much less significant. In this context, the initial high assignment error rate is primarily due to DeepClust cascaded clustering, whereby only representatives from each step are propagated to the next stage. As a result, a set of proteins that initially satisfy the clustering criterion with their representative may no longer do so after subsequent clustering steps, since their original representative is itself re-clustered with other proteins and replaced by a new higher-level representative. Consequently, not all members remain within the clustering criterion relative to the updated representative. In addition, the greedy vertex cover (GVC) algorithm used to partition the similarity network is equally responsible for the initial error rate, as it trades partitioning accuracy for computational efficiency by following a simple heuristic strategy where nodes with the highest degree are iteratively selected as representatives, grouped in a cluster with all their neighbors, and removed from the graph along with them. This causes early greedy choices to propagate suboptimal assignments and reduce clustering precision.

An initial attempt to replace GVC with MCL (see Section 2.2.3) revealed this consideration to be valuable in improving cluster quality at the cost of a significantly higher runtime. This was assessed through a comparative assessment on a sample of 500k proteins, comparing DIAMOND DeepClust with MCL applied on a similarity graph constructed from BLASTp ultra-sensitive searches with only the top five hits retained per sequence (-k 5). Multiple inflation parameter values were evaluated for MCL, ranging from 1.6 to 5.0, allowing explicit control over clustering

granularity, an advantage that does not apply to DeepClust. Notably, support for MCL-based clustering is also planned for integration in the next version of DIAMOND DeepClust [38].

The application of DeepClust at different sequence identity thresholds further highlighted its capacity to capture homology at multiple levels. Accordingly, DeepClust 80 could be regarded as a species-level partition. For our PanGBank-derived dataset, the sequences correspond to the representatives produced by a previous MMseqs clustering at 80% as part of PPanGGOLiN pangenome construction workflow. The further two-fold reduction in the total PanGBank representatives repertoire by DeepClust 80 reveals a significant proportion of highly similar universal prokaryotic families, which are likely involved in essential biological processes. On the other hand, DeepClust 50 and 0 could be regarded as genus-level and global partitions, respectively. This is partly supported by the proportions of intra- and inter-genus pairs produced by each level of clustering, as highlighted in Supplementary Fig. S8. In comparison, intra-species fractions were smaller than 1% across all partitions.

Lastly, ESM-based clustering presents a compelling alternative to alignment-based clustering, as it can capture remote structural and evolutionary relationships beyond explicit sequence similarity thresholds and as it proved to be highly promising based on our reported findings (Section 3.5.3). However, its application at full scale requires the proper computational resources, primarily in terms of GPU-based embedding computation, but also considering the costs of graph construction and community detection. Similarly, the nf-core/proteinfamilies workflow remains an interesting option for refined protein family construction, but its computational cost makes it challenging to apply to medium- and large-scale datasets such as PanGBank.

Annotation Annotation with different curated resources led to broad but incomplete and partially overlapping coverages, supporting retaining multiple annotation modalities in the final corpus. Coverage and redundancy of annotations per protein were also found to vary greatly across resources. Taken together, the different annotation layers provide complementary insights regarding PanGBank representatives, spanning conserved protein domains, functional classes, structural superfamilies, and orthologous groups. The integration of these insights will, therefore, enable the downstream LLM to learn patterns within each of them and across all of them simultaneously.

Embeddings The alignment-based homology patterns identified by DeepClust were largely preserved in the embedding space. Within-cluster cosine similarities were substantially higher than representative-to-representative similarities, indicating that proteins assigned to the same DeepClust cluster remain coherent under an independent ESM-based representation. However, the reverse was not always true: projection of DeepClust clusters into the embedding space showed that, although members of individual clusters formed coherent groups, several distinct clusters overlapped and could potentially be merged.

This observation provided an initial motivation for embedding-based clustering, which was subsequently applied to the final set of 4.5 million representatives obtained from the DeepClust 0 partition. At this level, no further alignment-based homology could be detected between representatives, making embeddings a useful alternative for exploring higher-order relationships. As reported above, the resulting embedding-based clusters displayed strong biological coherence, as supported by InterPro term enrichment and structural similarity analyses based on ESMFold.

Nevertheless, the ESM-based clustering strategy remains open to further refinement. Leiden clustering parameters should be optimized to better control supercluster granularity, while alternative community detection algorithms could also be explored. Similarly, the k-nearest-neighbor graph construction step could be varied or replaced by alternative graph-building strategies. Future work should also compare this approach with existing ESM-based clustering methods [40, 41]. In addition, structural validation was restricted to 3,000 sequences because of the computational cost of generating ESMFold structures; a more robust assessment would require extending this validation to a substantially larger scale.

The gradual transition from fully annotated to progressively less annotated regions in the PCA-projected embedding space (Fig. 13) warrants further exploration. This pattern is particularly notable because a similar distribution was observed for singleton proteins, and because shuffled ASTRAL40-derived sequences overlapped with poorly annotated regions of the embedding space. This resembles previous observations linking such regions to sequences with low structural prediction confidence [64], suggesting that these areas may correspond to poorly characterized, compositionally unusual, or structurally uncertain proteins.

The demonstrated ability of ESM embeddings to group representatives with no detectable sequence similarity but shared annotations provides a direct basis for annotation transfer. More broadly, this suggests a potential shift beyond the traditional annotation paradigm based primarily on sequence alignment and HMM profiles, toward embedding-based strategies capable of capturing remote structural, evolutionary, and functional relationships. Such approaches may provide a powerful avenue for annotating poorly characterized proteins. It also offers a solution to handle PanGBank proteins left unannotated by any tool, which pose an important challenge for downstream LLM training.

Overall, ESM embeddings proved their capacity to provide an independent and complementary representation of the PanGBank protein space. They supported the consistency of DeepClust clusters after correcting embedding anisotropy, revealed structured transitions between annotated and poorly annotated regions, and enabled construction of broader superclusters that summarize large-scale organization of representative proteins.

Tokenization Rather than training a graph-based model directly on PPanGGOLiN pangenome graphs, annotations are propagated to all constituent genomes to construct a large sequential corpus for LLM training. The graph structure inferred by PPanGGOLiN through Neighboring Expectation-Maximization (NEM) partitioning should be learned by the LLM. This design avoids the added complexity of graph-based architectures and allows the LLM to learn functional and evolutionary patterns from the linear organization of genes along genomes. Accordingly, tokenization is performed at the level of annotations, with genomes represented as ordered sequences of annotation tokens along each contig. A sliding window over contigs is then used to generate training segments, treating each window as a sentence- or paragraph-like unit. Moreover, annotation coverage improved substantially after propagating representative-level annotations back to genome-level proteins. In particular, the proportion of unannotated proteins, referred to as "dark matter" was reduced to less than 5% in the final corpus, while coverage by individual annotation resources also increased significantly.

However, the annotation propagation revealed great lack of uniformity in the sizes of PanGBank pangenomes, with a small number of species represented by many more genomes and

therefore contributing disproportionately more annotation tokens. This size imbalance, together with differences in annotation coverage across tools, could bias the model toward over-represented taxa or well-annotated functional spaces. In addition, genome lengths vary substantially, which may make direct genome-level tokenization inefficient because of excessive padding or truncation. A more suitable strategy could rely on biologically informed sliding windows, for example based on gene neighborhoods, contigs, operon-like structures, or fixed-size genomic regions.

In addition, while most proteins were associated with a small number of terms per resource, with a median of one annotation across all vocabularies, multi-token proteins were observed for all resources, with maximum values ranging from 11 terms per protein for SUPERFAMILY to 16 for Pfam. This supports a tokenization strategy capable of representing multiple annotations per protein rather than forcing a single-label representation. Such ordered multi-token encoding is particularly important for multi-domain proteins identified with Pfam. To this end, a special gene separator token was proposed to allow for multiple annotation terms between separators.

Vocabulary design is another central issue. One possibility is to treat each annotation resource as a distinct language, allowing the model to learn translation-like relationships between modalities and to predict missing annotations from the modalities that are available. This would preserve tool-specific biological information but would require a more complex multimodal architecture. Conversely, merging all annotations into a unified vocabulary would simplify the model and help overcome uneven annotation coverage, but risks mixing biologically distinct annotation types. A hybrid strategy may therefore be most appropriate: closely related resources could be merged to reduce redundancy and improve coverage, while functionally or structurally distinct modalities remain partially separate.

Vocabulary optimization could also be further explored beyond raw annotation terms. The DeepClust-based grouping of Pfam terms already suggests that clustering can refine the vocabulary toward more language-like frequency structure, and a similar strategy could be tested using ESM-based embedding neighborhoods or superclusters. Future work should also repeat the Zipf-law, Gini, and vocabulary-frequency analyses at the all-genome level after annotation propagation, and extend them with additional corpus diagnostics such as token co-occurrence, modality agreement, rare-token retention, and taxonomic balance.

Limitations There are also several notable limitations. First, the annotation corpus remains dependent on the coverage, quality, and versioning of external databases. EggNOG-mapper, for example, remains not maintained since 2019, which requires running it against the eggNOG database v5.0.2, while newer database releases have significantly improved annotation breadth and consistency, the latest of which eggNOG v7 was released this year [69].

Second, annotation propagation relies on the assumption that representative protein families capture functionally coherent groups. Although the clustering validation analyses support this assumption overall, some clusters may still contain functionally divergent members. Notably, even DeepClust 80 displayed a member-to-representative annotation mismatch rate higher than 10% for some tools. Considering that all PanGBank representatives are obtained with MMseqs at 80%, and assuming comparable accuracy, this will mean that up to 10% of the 1 billion propagated annotations will be inaccurate, corresponding to up to 100 million proteins.

Third, the corpus is affected by dataset imbalance. A small number of pangenomes contain many more genomes and therefore contribute disproportionately more tokens to the final training

corpus, while rare taxa and poorly annotated protein families remain under-represented. Without explicit balancing, downsampling, or loss reweighting, downstream models may preferentially learn patterns from highly represented taxa and well-characterized functional spaces.

Finally, embedding-based analyses were limited by the choice of model, pooling strategy, normalization, and computational constraints. ESM embeddings provided useful complementary structure, but only a subset of possible protein language models and embedding transformations was explored. Similarly, the proposed tokenization strategies were evaluated through corpus-level diagnostics, but downstream LLM training and task-specific benchmarking remain necessary to determine which vocabulary design, modality structure, and missing-annotation handling strategy performs best in practice.

5 Conclusion

In conclusion, this work established a large-scale workflow for transforming PanGBank representative proteins into a structured annotation corpus suitable for downstream LLM training. The first step consisted of clustering representative proteins across PanGBank pangenomes, where DIAMOND DeepClust provided a scalable and complete partition of the 34.6 million representative sequences. Criterion-based, annotation-based, and embedding-based validation then showed that these clusters captured substantial biological consistency, while refinement through reclustering and reassignment improved member-to-representative coherence and produced a more reliable representative set.

Subsequently, a multimodal annotation workflow was developed and applied across representative proteins using complementary functional, structural, and orthology-based resources. The resulting annotations showed broad but incomplete coverage, with each tool contributing partially overlapping information. Functional consistency analyses further supported the biological coherence of the refined sequence-based clusters, while also highlighting the value of retaining multiple annotation modalities rather than relying on a single resource.

Protein embeddings provided an additional layer of representation and validation. After correcting the anisotropy of raw ESM embeddings, cluster members were shown to remain closer to their assigned representatives than expected from representative-level background similarity. Embedding-space projections further revealed structured transitions between well-annotated and poorly annotated regions, suggesting that embeddings capture information complementary to curated annotations. Finally, ESM-based superclustering further reduced the representative protein space into broader groups, whose functional enrichment and structural validation supported the biological relevance of the embedding-derived partitions.

The final stage propagated representative-level annotations back to the proteins of the constituent genomes of each pangenome, expanding the corpus from millions of representative proteins to over one billion protein occurrences. This enabled the construction and analysis of annotation vocabularies for tokenization, revealing long-tailed frequency distributions, strong token imbalance, multi-token proteins, and modality-specific sparsity patterns. Vocabulary optimization strategies, including DeepClust-based grouping, provided a promising route toward more compact and natural language-like token spaces. Overall, this work provides a foundation for training an annotation-level LLM for prokaryotic pangenomes.

Code and Data Availability

Computed clusters, annotations, and embeddings will be integrated into PanGBank (https://pangbank.genoscope.cns.fr/).

The annotation workflow is available at https://github.com/labgem/PanGBank-annotator. Scripts used for clustering, annotation, embedding computation, tokenization, results analysis, and plotting will be made available at https://github.com/yazid-hoblos/panCATE.

AI Use Statement

AI models, primarily GPT-5.5, were used to assist with code development and the preparation of the summary diagrams in Figures 1, 2, and 3, and Supplementary Figures S2 and S3. All outputs were reviewed, validated, and edited by the author.

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A Supplementary Figures

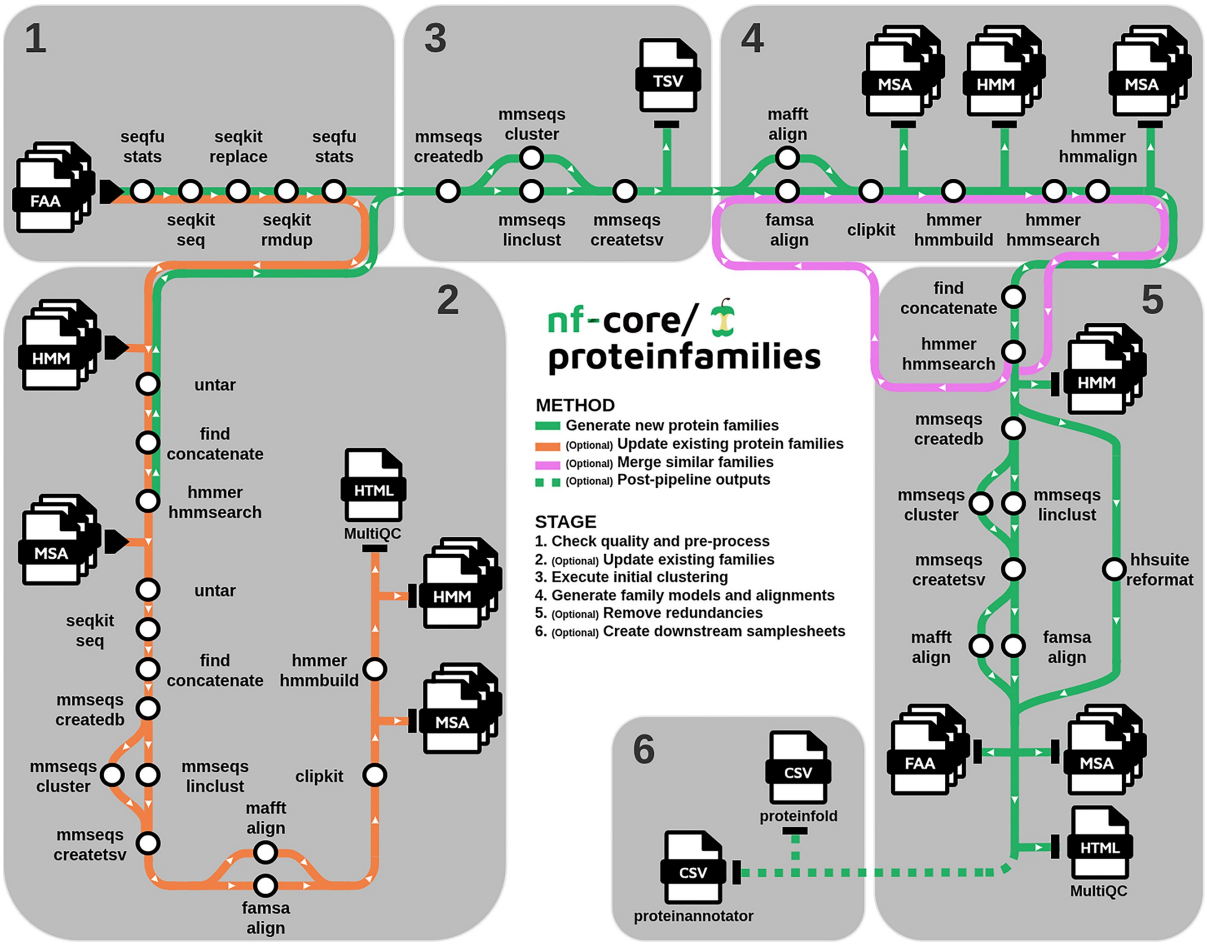


Figure S1: Schematic overview of the nf-core/proteinfamilies workflow, detailing the transition from alignment-based MMseqs2 clustering to profile-based HMM refinement [39].

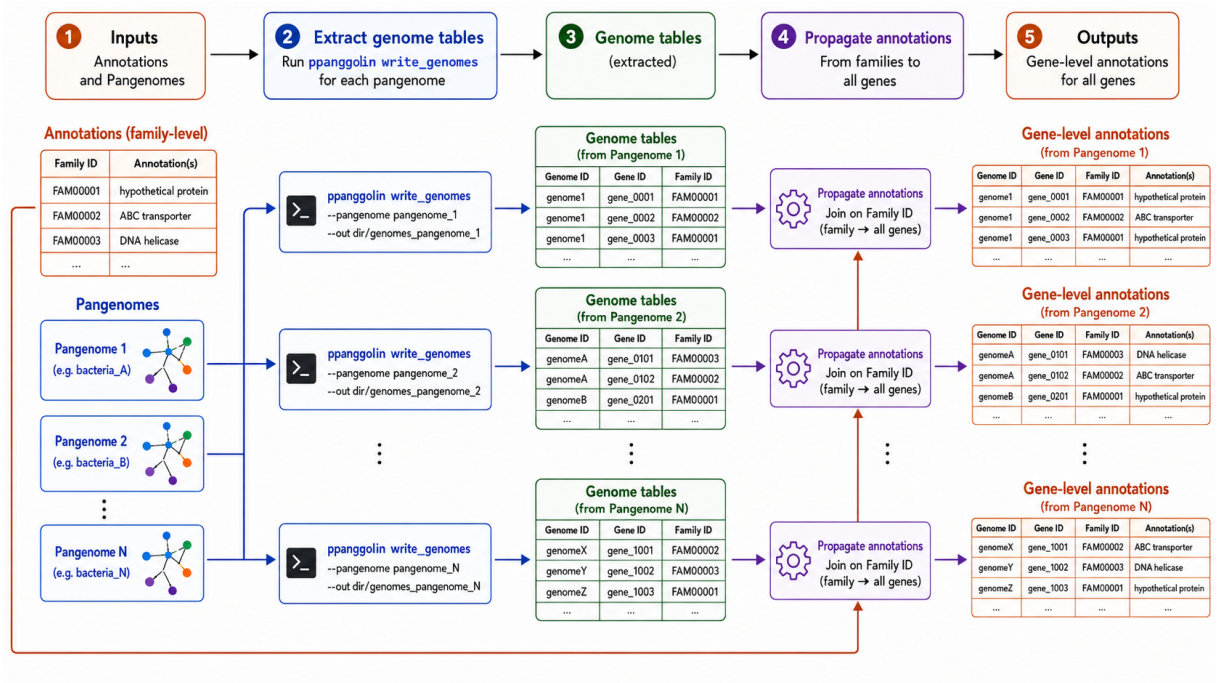


Figure S2: Back-propagation of representative-family annotations to genome-level gene occurrences for tokenizer construction. Functional annotations inferred on representative proteins are mapped back to their corresponding pangenome families and genomic contexts, enabling ordered annotation sequences to be generated for downstream language-model training.

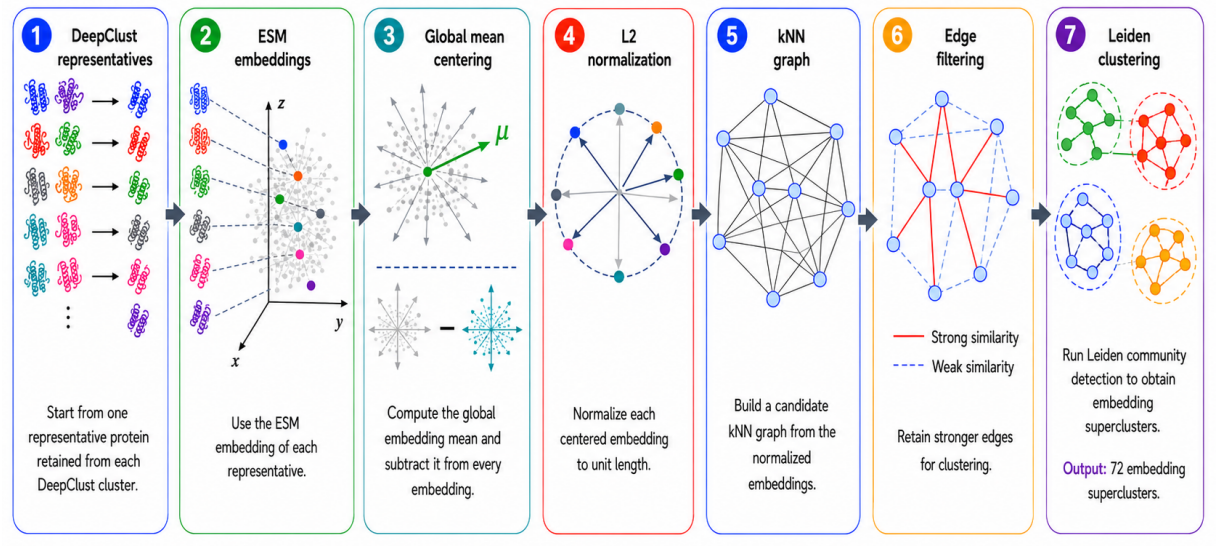


Figure S3: Schematic overview of the ESM-based superclustering workflow. Representative proteins are embedded in ESM space, grouped into higher-level superclusters, and analyzed to assess large-scale organization beyond alignment-based clustering.

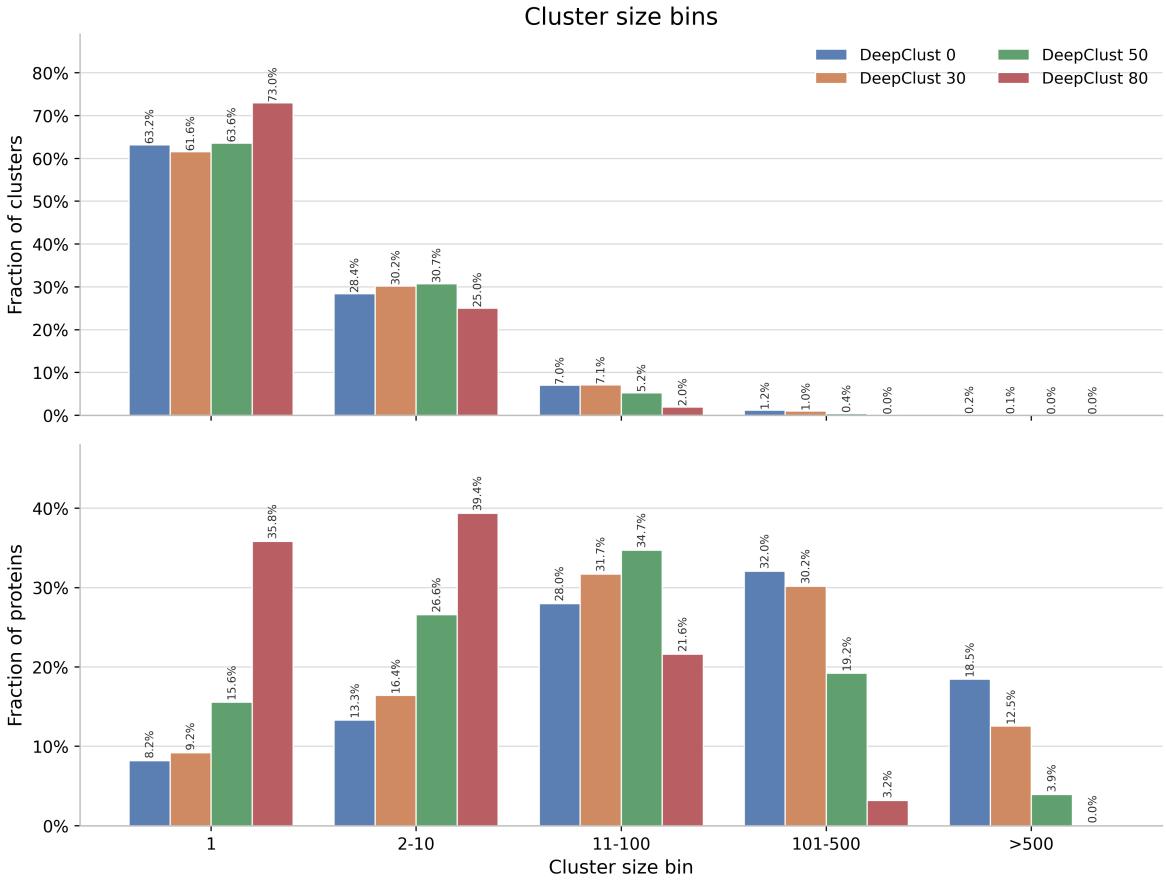


Figure S4: Binned cluster-size distributions across DeepClust partitions. Top: fraction of clusters in each size bin. Bottom: fraction of proteins in each cluster-size bin.

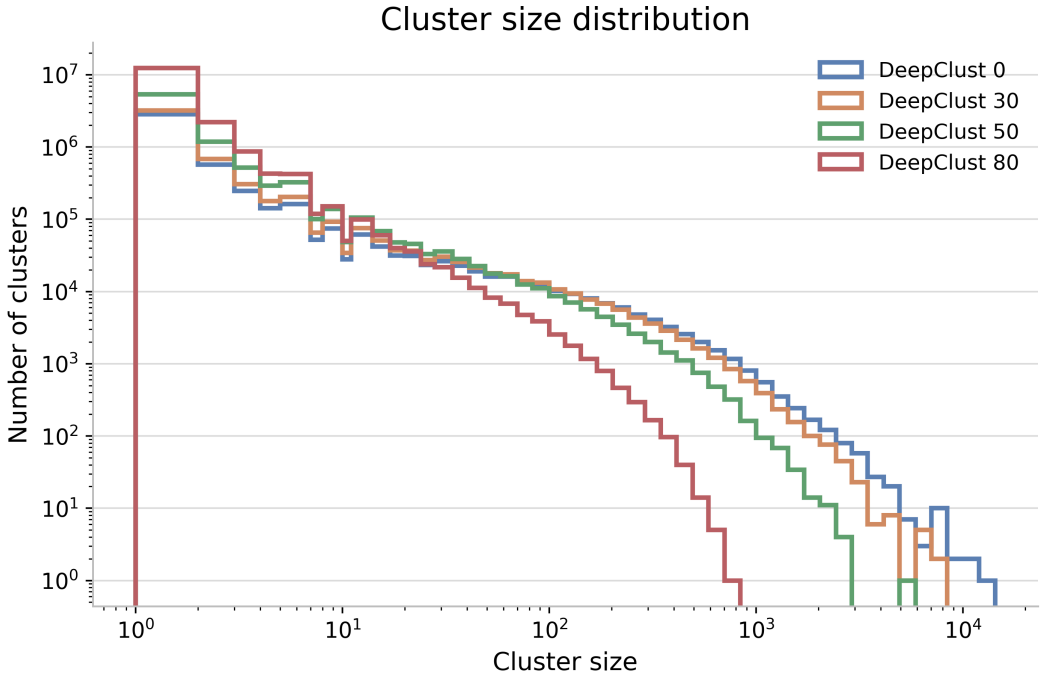


Figure S5: Cluster-size distributions across DeepClust partitions. The distributions are similar for small cluster sizes but diverge in the tail, with lower stringency partitions showing a slower decrease and larger maximum cluster sizes.

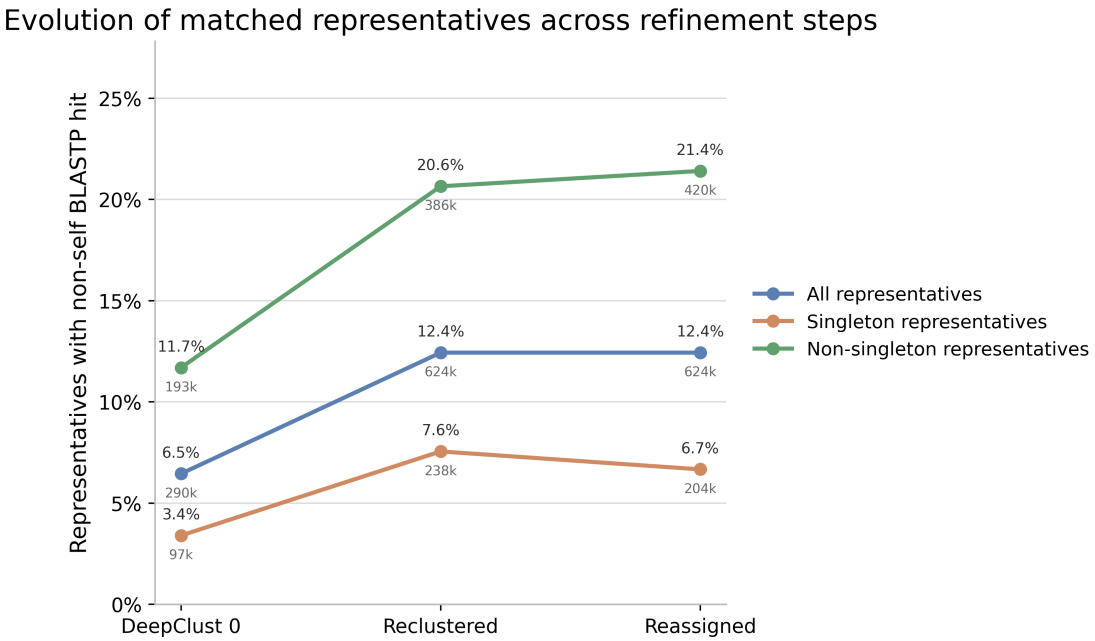


Figure S6: Evolution of representative-level non-self BLASTp matches across refinement steps. Reclustering increases the number of representatives with non-self hits by splitting broad heterogeneous clusters into finer representative groups, while reassignment preserves the representative set but redistributes matches from singleton toward non-singleton representatives.

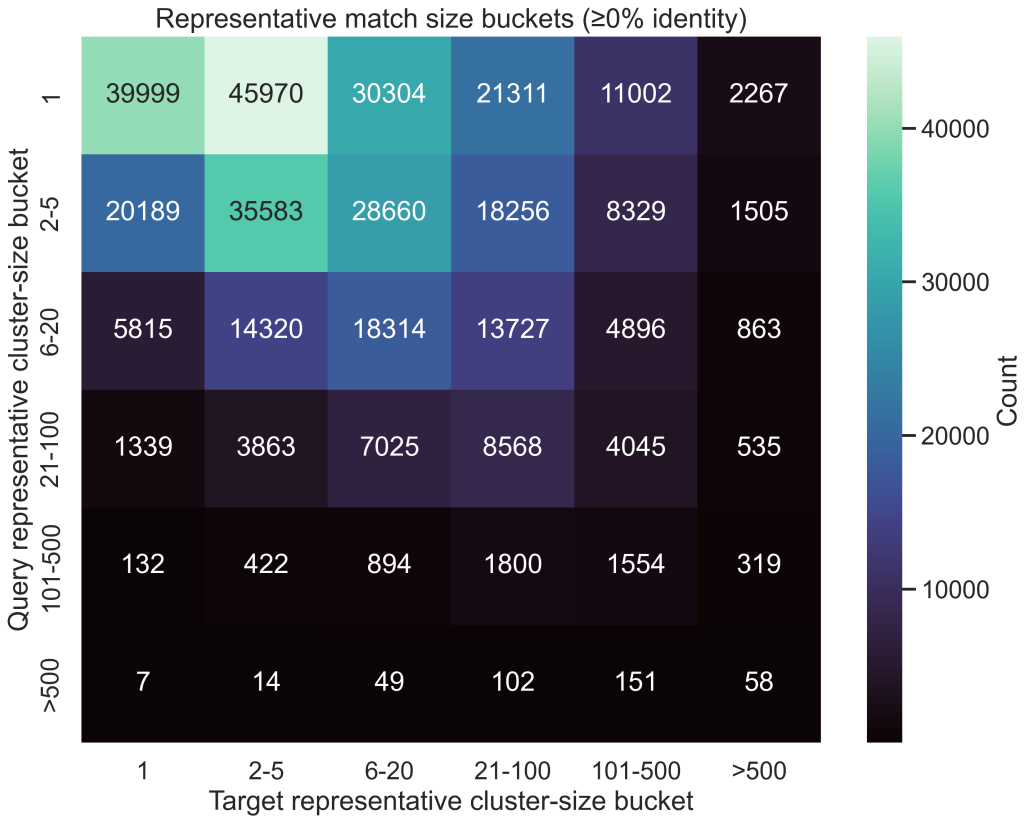


Figure S7: Cluster-size distribution of residual assignment inconsistencies after refinement. Size-bucket diagnostics show that remaining inconsistencies are concentrated in specific cluster-size classes rather than uniformly distributed across the clustering.

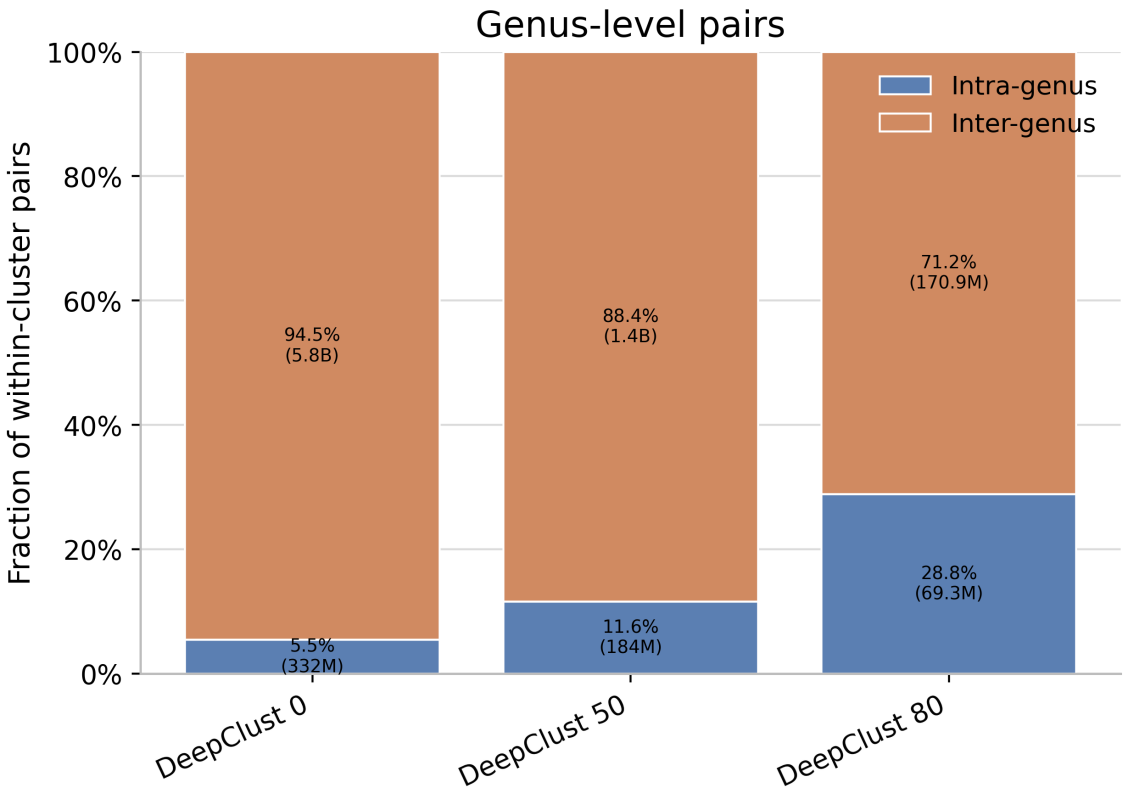
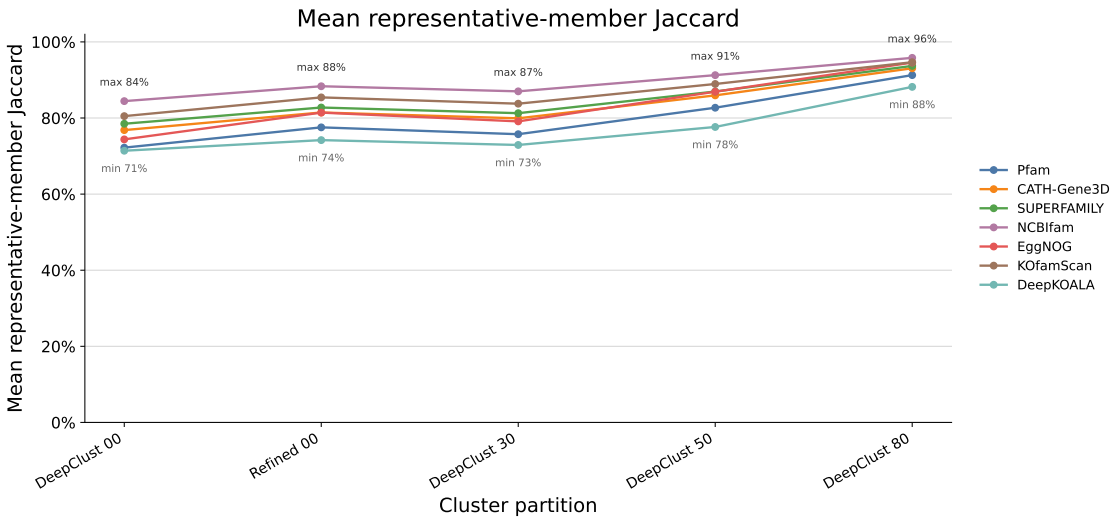
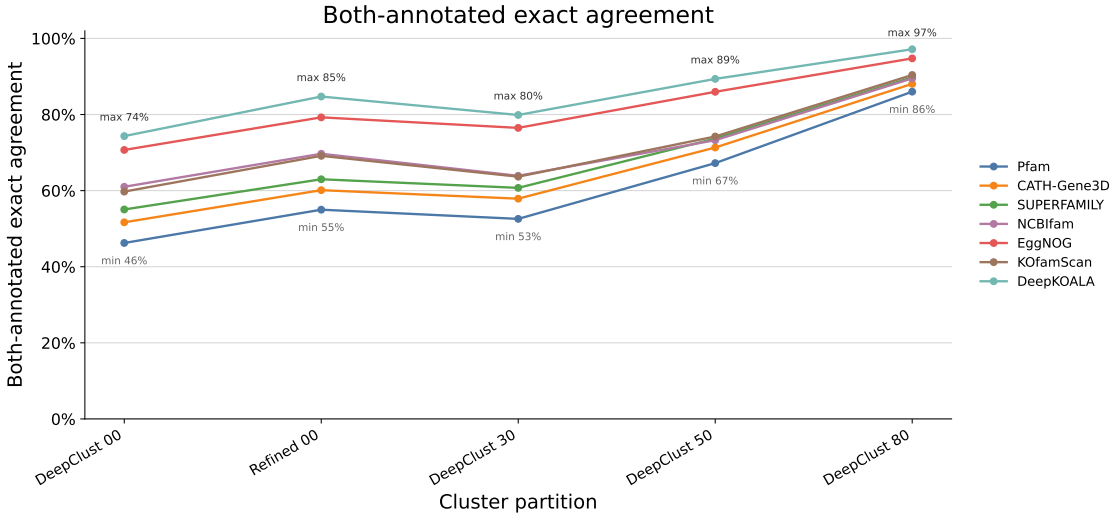


Figure S8: Fractions of within-genus and between-genus protein pairs across DeepClust partitions. Increasing clustering stringency raises the proportion of within-genus pairs, while between-genus relationships remain predominant. Labels indicate the corresponding pair counts.



(a) Mean annotation-set Jaccard similarity across all representative–member pairs.



(b) Exact annotation-set agreement among representative–member pairs for which both proteins were annotated.

Figure S9: Functional consistency of alignment-based clusters across annotation resources and clustering partitions. Jaccard similarity measures partial overlap between annotation sets, whereas exact agreement requires identical annotation sets for the representative and member.

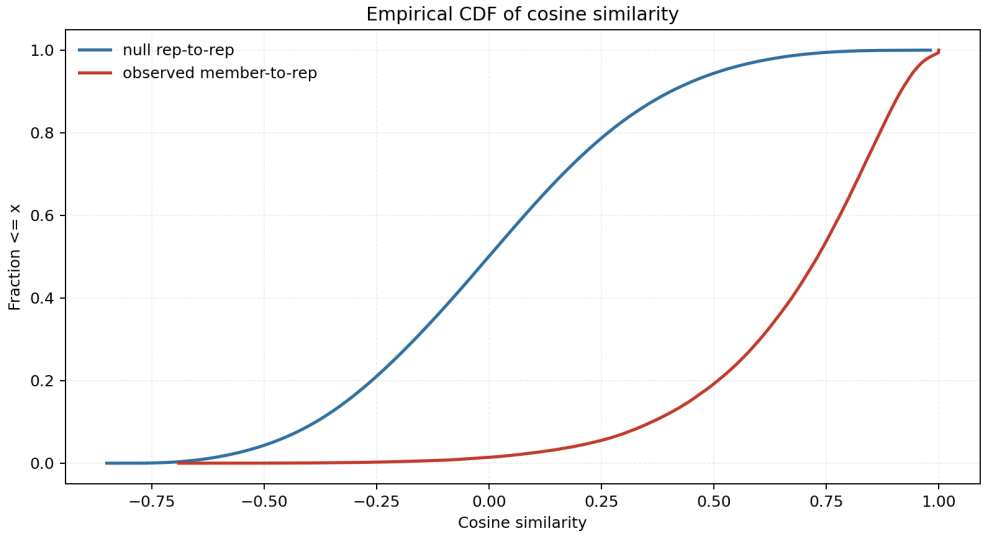


Figure S10: Cumulative distribution functions comparing centered member-to-representative cosine similarities with the representative-to-representative null distribution.

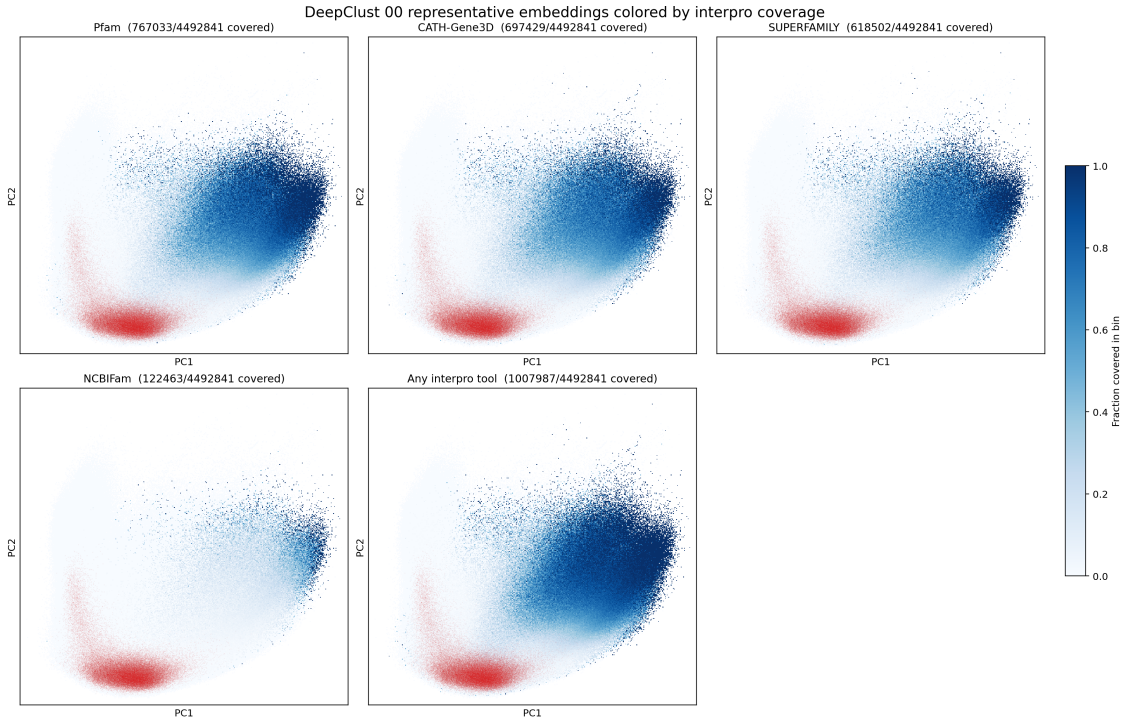


Figure S11: Organization of InterProScan annotation coverage in the ESM embedding space. Representative proteins were projected into two dimensions and colored according to annotation coverage across Pfam, CATH-Gene3D, SUPERFAMILY, NCBIfam, and any InterProScan resource. Shuffled ASTRAL40-derived sequences were added as random controls. The overlap between random controls and part of the poorly annotated space helps distinguish composition-driven regions from potentially structured but under-characterized protein families.

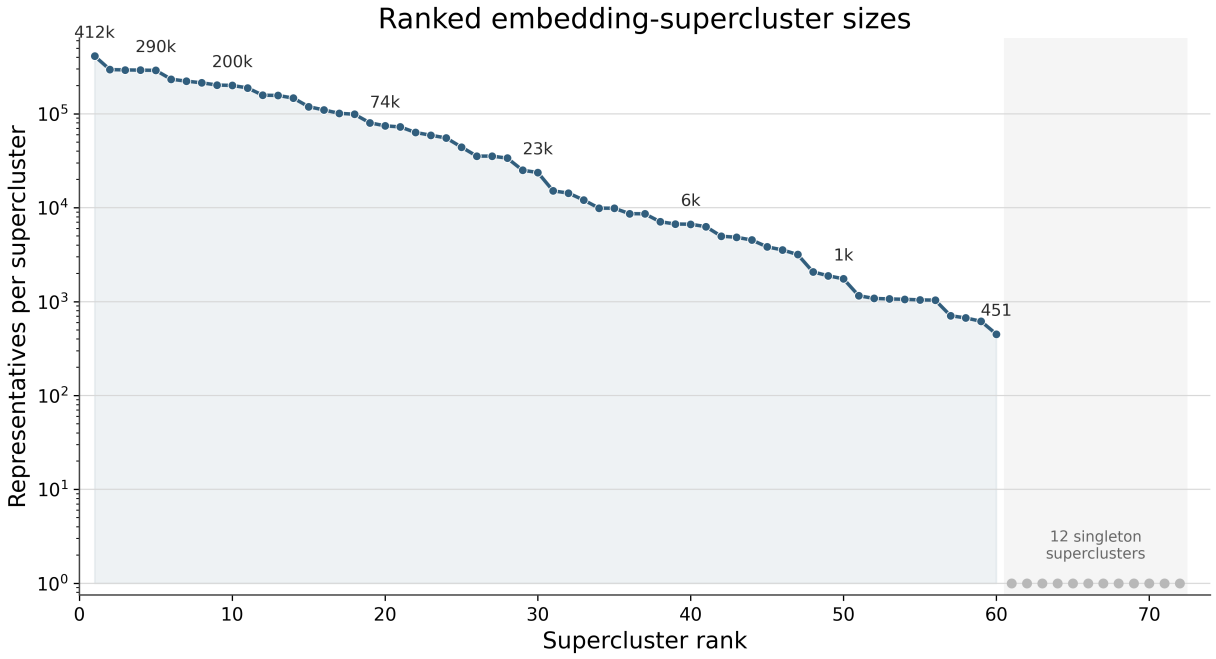
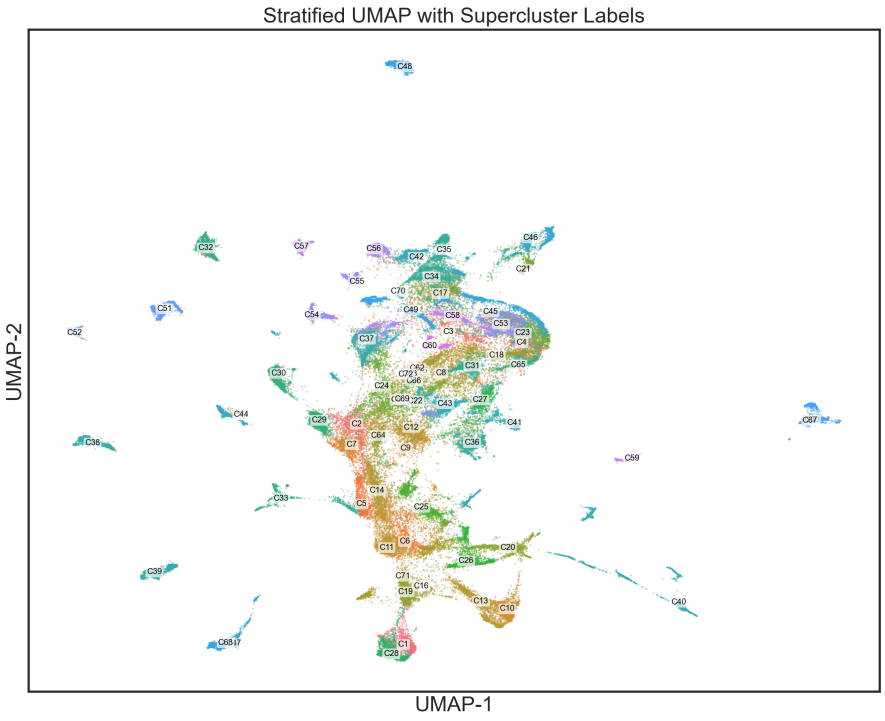
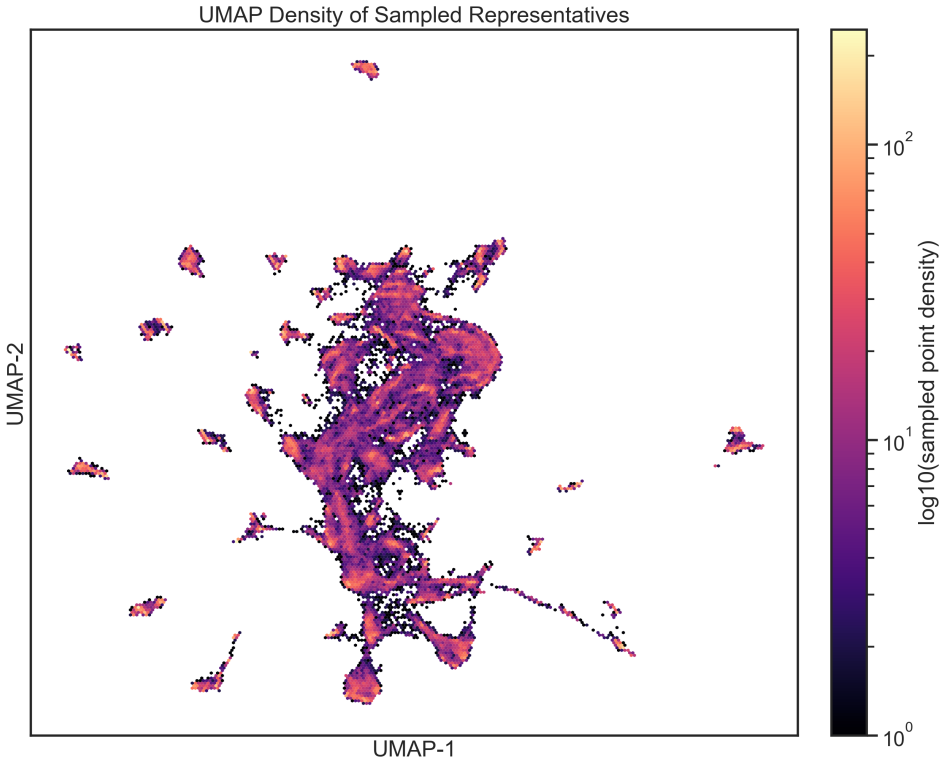


Figure S12: Ranked size distribution of ESM-based superclusters. A limited number of large superclusters account for most protein representatives, while several singleton superclusters remain at the tail of the distribution.



(a) Sampled representatives colored and labeled by ESM-based supercluster.



(b) Density of representatives across the same projected embedding space.

Figure S13: Organization of ESM-based superclusters in the protein embedding landscape. The labeled projection shows the spatial distribution of superclusters, while the density view highlights densely and sparsely populated regions.

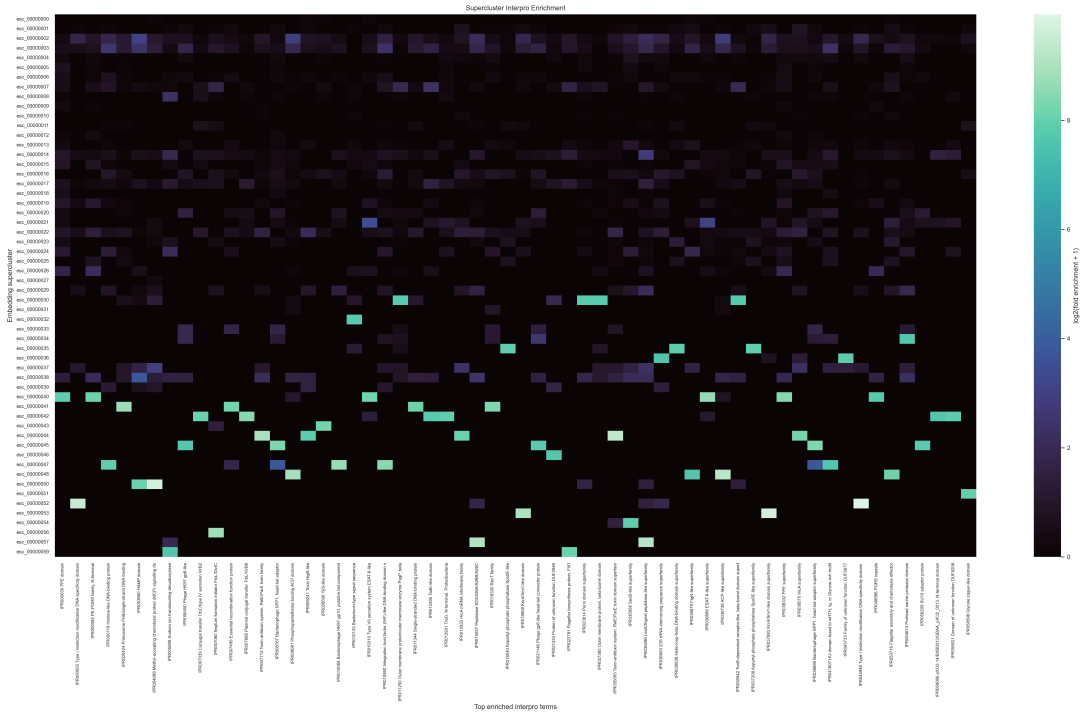


Figure S14: InterProScan annotation enrichment across ESM-based superclusters. Distinct enrichment profiles indicate that superclusters capture broad functional and structural organization in the protein embedding space.

Table S1: Top significant InterPro enrichment for each ESM-based supercluster. Significance was assessed using a Bonferroni-corrected threshold of $p < 0.05/77,787$.

SC	Top enriched InterPro term	Observed	Frequency	Fold
52	Type I restriction modification DNA specificity domain superfamily	728	68.1%	981.5
50	Methyl-accepting chemotaxis protein (MCP) signalling domain	578	50.0%	832.6
53	Knr4/Smi1-like domain superfamily	568	53.6%	815.4
44	Toxin-antitoxin system, RelE/ParE toxin domain superfamily	1,002	26.1%	646.7
48	ACP-like superfamily	776	41.1%	599.0
57	LexA/Signal peptidase-like superfamily	393	58.6%	558.3
56	Septum formation initiator FtsL/DivIC	245	34.6%	443.4
41	Primosome PriB/single-strand DNA-binding	548	11.0%	432.2
47	Bacteriophage HK97-gp10, putative tail-component	255	12.3%	403.6
40	ESAT-6-like superfamily	627	10.0%	392.9
42	Plasmid conjugal transfer TrbL/VirB6	299	6.2%	353.7
45	Bacteriophage SPP1, head-tail adaptor superfamily	237	6.7%	326.6
43	YjiS-like domain	214	4.7%	289.0
59	Flagellar biosynthesis protein, FliO	141	31.3%	272.1
51	Glycine zipper-like domain	134	12.4%	251.3
54	SecB-like superfamily	128	12.3%	236.1
46	Protein of unknown function DUF2946	120	3.8%	203.1
55	YjzH-like	55	5.3%	106.6

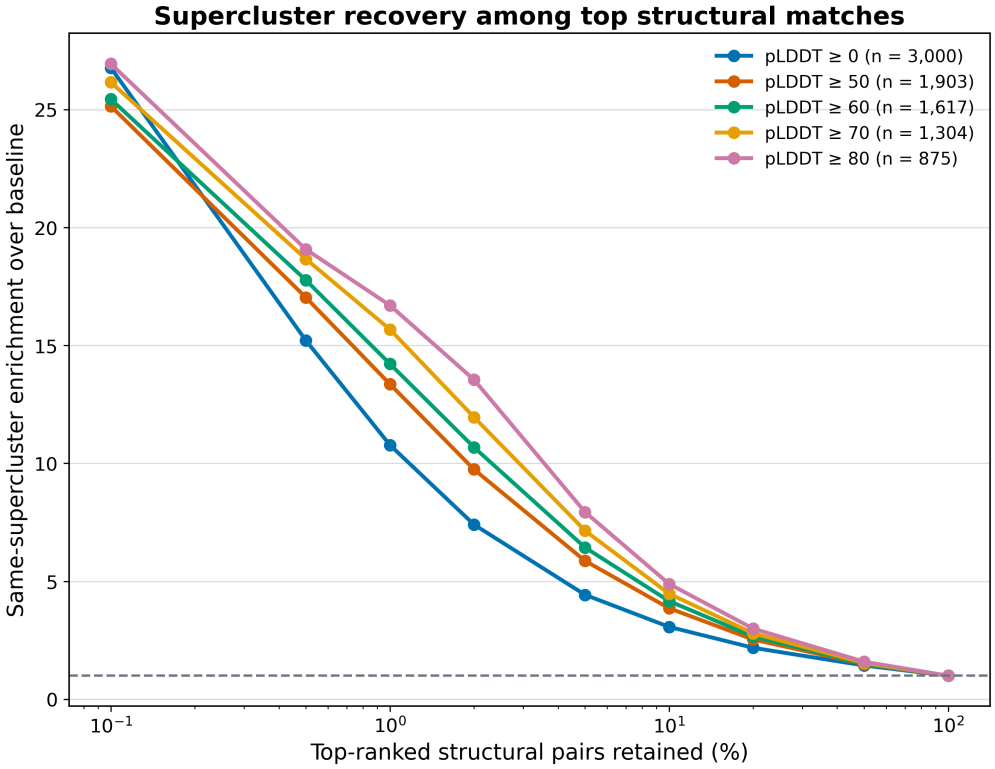
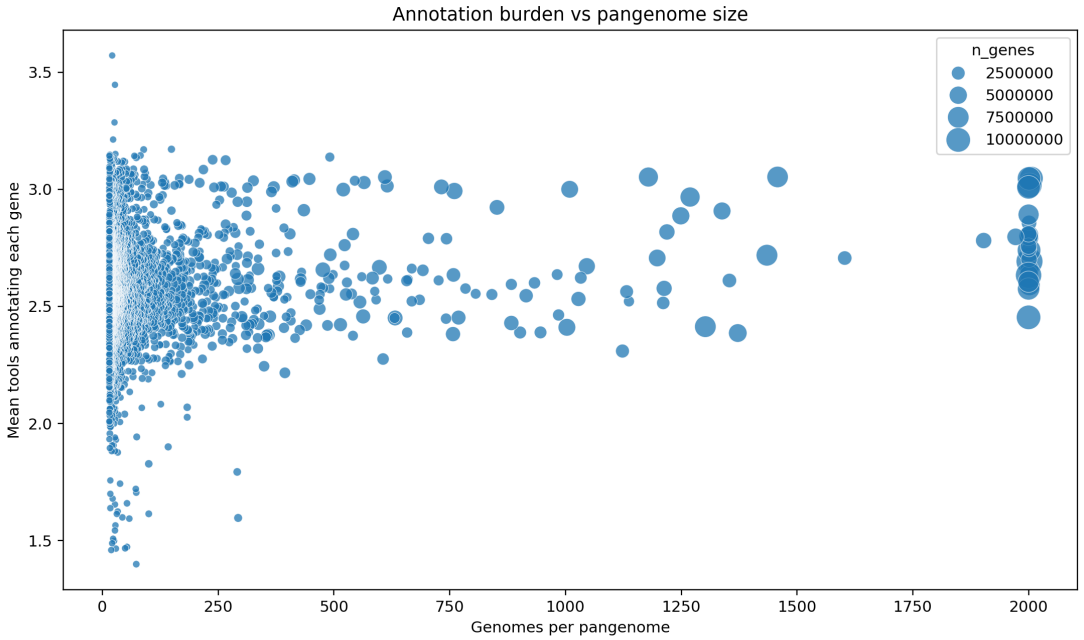
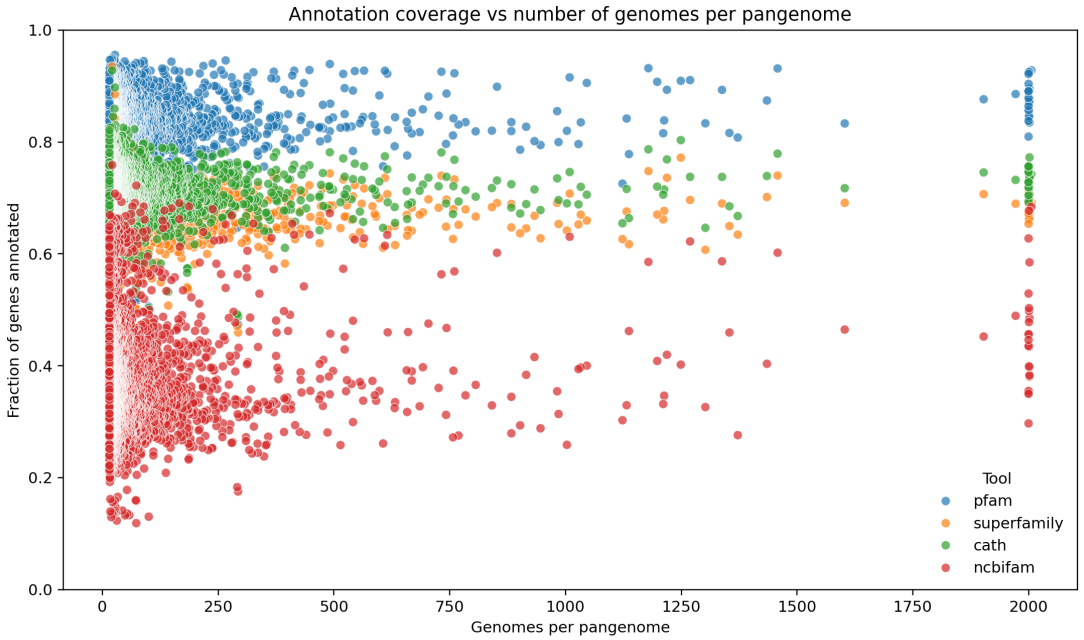


Figure S15: Enrichment of same-supercluster pairs among the top-ranked structural matches. Structural pairs were ranked by conservative Foldseek similarity, and same-supercluster precision was compared against the prevalence baseline.



(a) Mean number of annotation tools per gene across pangenome sizes. Point size reflects the total number of genes in each pangenome.



(b) Per-tool annotation coverage across pangenome sizes.

Figure S16: Relationship between pangenome size, per-gene annotation support, and per-tool annotation coverage after propagation to genome-level genes.

B Clustering approaches benchmark

A detailed comparison of DIAMOND DeepClust, MMseqs2, and the nf-core/proteinfamilies workflow was performed on a subset of 400k representative sequences, selected randomly from the total repertoire of all PanGBank GTDB-all representatives (see Table 1). This benchmark assessed not only the number of clusters produced by each approach, but also the extent to which the resulting partitions agreed with one another. In addition, as the nf-core/proteinfamilies workflow generated significantly fewer and larger clusters, the extent to which its final families behaved as higher-level aggregates of raw sequence clusters was assessed. This is further justified by the internal use of MMseqs clustering within the nf-core workflow.

On the 397,868 input proteins, DeepClust and MMseqs both produced complete non-overlapping partitions, whereas the final nf-core/proteinfamilies family table covered only 94,910 proteins after its HMM-based refinement, family merging, redundancy removal, and minimum-size filtering steps (Supplementary Fig. S1), as presented in Table S2. DeepClust produced 153.5k clusters, compared with 189k clusters for both standalone MMseqs and the initial MMseqs step internal to nf-core/proteinfamilies. The singletons fraction was also notably lower for DeepClust.

Method	Members	Clusters	Singletons	Median size	Max size
DeepClust 0%	397,868	153,683	106,836	1	373
MMseqs 0%	397,868	188,926	145,236	1	639
nf-core initial MMseqs	397,868	189,006	145,318	1	561
nf-core final families	94,910	1,050	0	47	4,150

Table S2: Summary of clustering outputs on the random benchmark sample. The nf-core final family output is not directly equivalent to the raw sequence clusterings, as it represents a post-processed family catalogue after profile-based refinement and filtering.

Directional overlap analysis further showed that most clusters had a direct counterpart in the other method. Yet, while from DeepClust to MMseqs, 90% of clusters had a best-match Jaccard index of at least 0.5 and 77% reached at least 0.8, only 76% of MMseqs clusters reached a best-match Jaccard index of 0.5 and 62% reached 0.8, in the reverse direction. This asymmetry is consistent with MMseqs producing a more fragmented partition, while DeepClust grouped more proteins into fewer clusters.

The global comparison between DeepClust and MMseqs showed substantial agreement despite their different fragmentation levels (Fig. S17). Across all 397,868 shared proteins, the adjusted Rand index (ARI) was 0.47, while the normalized mutual information (NMI) was 0.96 (see Section 2.2.1 for their definitions). This combination of a moderate ARI and high NMI indicates that while the two methods recover broadly similar clustering structure, they differ in the exact pairwise grouping of proteins. In particular, the high NMI suggests that knowing the cluster assignment from one method is highly informative about the assignment from the other, whereas the moderate ARI reflects differences in cluster granularity and fragmentation.

The final nf-core/proteinfamilies output showed a different structure from the raw sequence partitions. After normalization of domain-like member identifiers to their base protein identifiers, the final 1,050 nf-core families covered 24% of the input proteins. These final families frequently aggregated several raw clusters rather than corresponding to a single DeepClust or MMseqs cluster

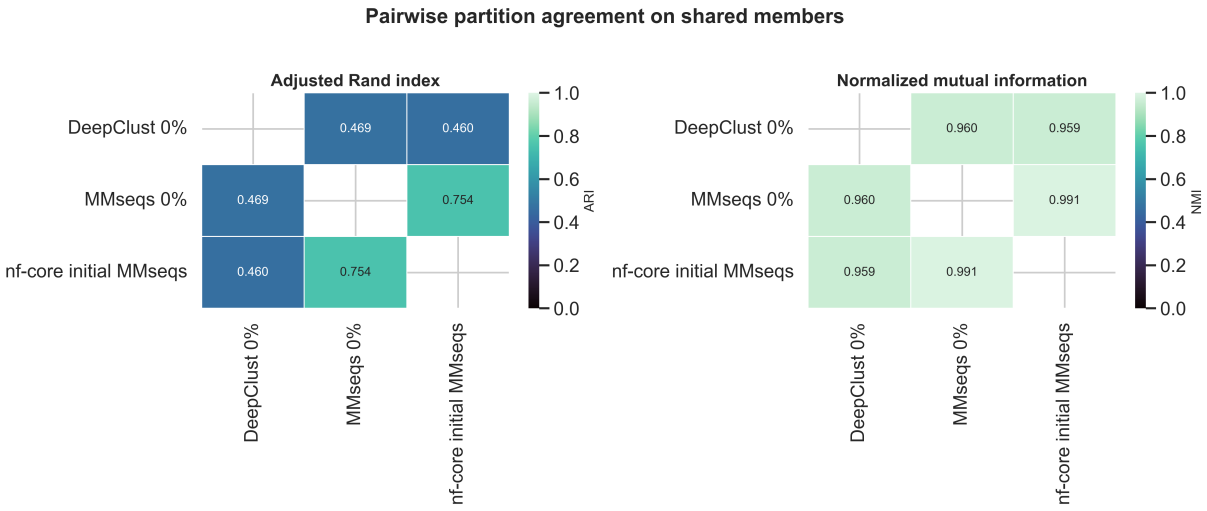


Figure S17: Pairwise agreement between raw clustering partitions on the random 1% benchmark sample. Agreement was measured using adjusted Rand index and normalized mutual information on proteins shared by each pair of methods.

(Fig. S18). Relative to DeepClust, the median nf-core final family touched four raw clusters, while 48% touched at least five DeepClust clusters and 11.6% touched at least 20. Relative to MMseqs, the median final family touched five raw clusters, 51% touched at least five MMseqs clusters, and 13.4% touched at least 20. This confirms that the final nf-core/proteinfamilies output is better interpreted as a refined higher-level family catalogue than as a complete raw clustering partition.

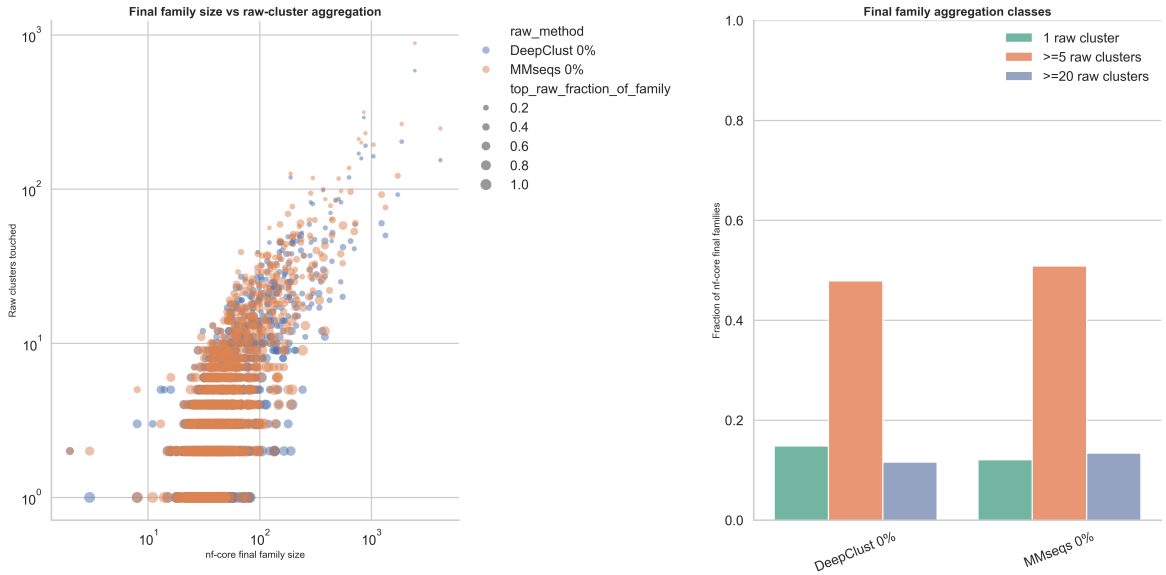


Figure S18: Relationship between nf-core final family size and the number of raw clusters touched in DeepClust and MMseqs. Points represent nf-core final families, colored by the raw method being compared. Larger values on the y-axis indicate final families that aggregate multiple raw clusters.

Finally, co-clustering agreement was evaluated using protein pairs as the unit of comparison (Fig. S19). A pair was counted for a method when both proteins were assigned to the same cluster by that method. On the 94,910 proteins shared with the normalized nf-core final family output, 1,104,241 protein pairs were co-clustered by all three approaches. In contrast, 22.5k pairs were co-clustered by DeepClust and MMseqs only, 632k by DeepClust and nf-core only, 1.3M by

MMseqs and nf-core only, and 21.6M by nf-core only. The large nf-core-only region reflects the broader family aggregation behavior described above.

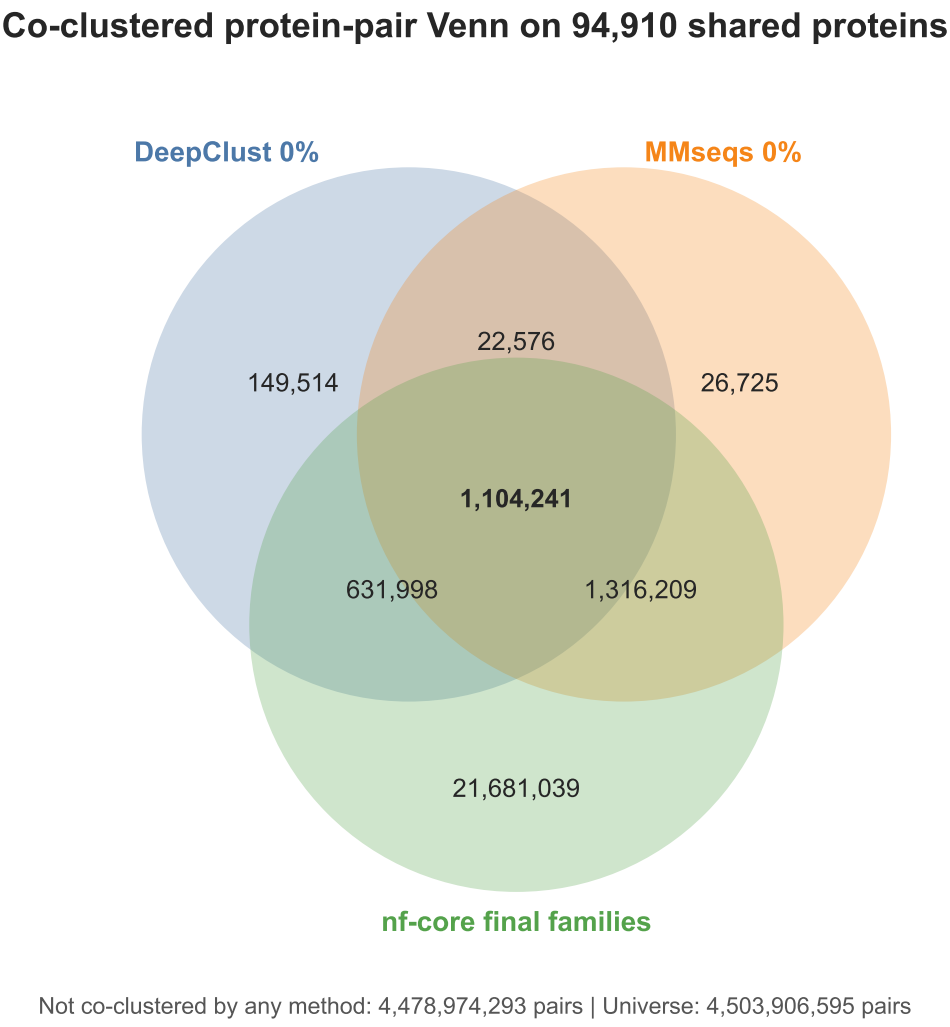


Figure S19: Protein-pair co-clustering Venn diagram for DeepClust, MMseqs, and final nf-core/proteinfamilies families. The universe corresponds to protein pairs among the 94,910 proteins represented in the normalized nf-core final family output.

Overall, these results motivated the use of DeepClust as the primary large-scale clustering method for PanGBank. It preserved complete protein coverage, produced fewer clusters and fewer singletons than MMseqs, and retained strong agreement with MMseqs for most cluster assignments. In contrast, nf-core/proteinfamilies provided biologically useful refined families but did not produce a complete one-to-one partition of all input proteins, making it less suitable as the sole clustering layer for annotation transfer, representative selection, and corpus construction. Furthermore, while DeepClust displayed a runtime in the range of a few hours at the scale of millions of sequences, nf-core/proteinfamilies required days or even weeks at comparable scales.